

K-FIELD LUCAS SUPERGROUP LARGE-SCALE MORPHOLOGY

Exact volume-filling extension of the fully defined 9B Lucas Supergroup: general cuboidal laws, cubic growth, recursive closure, boundary-to-bulk transition, sheet and shell morphology, reciprocal spectra, and emergent multiscale structure

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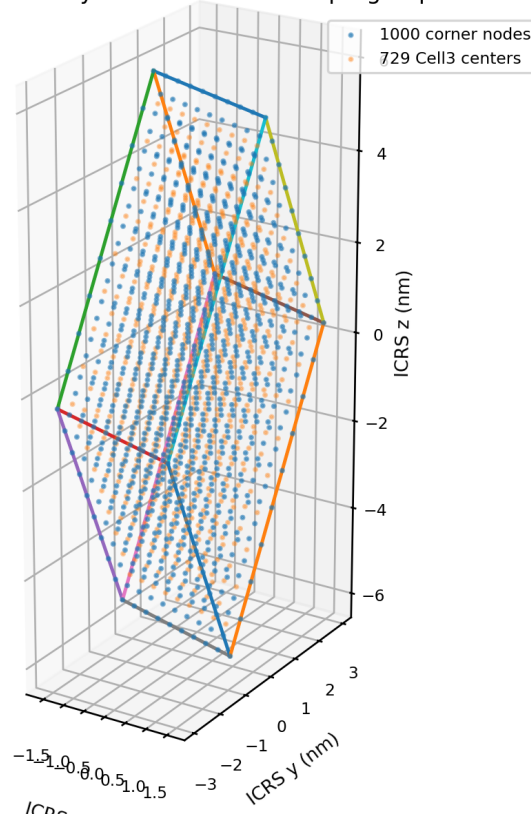
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Dimensionally correct 9B Lucas Supergroup



Title figure. Dimensionally correct ICRS rendering of the 9B Lucas Supergroup. The exact block contains 729 Cell3 volumes, 27 Lucas Groups, 1,000 merged corner nodes, and 729 Cell3 centers. Every plotted coordinate is generated from the authoritative Cell3 basis B.

Archival computational edition - Unix epoch 1785242330

Document Control

Field	Value
Title	K-Field Lucas Supergroup Large-Scale Morphology
Release type	Foundational educational and archival computational reference
Principal Investigator	P. Dwayne Esterline
Credentials	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC
Epoch identifier	1785242330
UTC generation time	2026-07-28T12:38:50Z
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Authoritative source	K-Field_Lucas_Supergroup_extracted_definition_1785238606(1).json
Authoritative source SHA-256	79b8aadb50be81da2569a901063db166fd5710377d644412638b25ffacd4526a
Coordinate frame	ICRS Cartesian; columns of B are the authoritative direct basis vectors
Units	nm, nm ² , nm ³ , and reciprocal nm ⁻² unless stated otherwise
Machine-readable companion	K-Field_Lucas_Supergroup_large_scale_morphology1785242330.json
PDF generation	Python ReportLab; built-in core fonts only

The term Lucas Supergroup denotes the exact 9B oblique cubical complex containing 729 Cell3 volumes, equivalently 27 Lucas Groups. This report extends that object into aligned volume-filling assemblies without altering its locked basis, node system, plane geometry, polarity grading, or reciprocal metric.

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Abstract

A fully defined Lucas Supergroup provides an exact building block for constructing larger filled volumes. The extension is governed by the same oblique basis B that defines Cell3, so scale growth changes occupancy and topology while preserving the underlying metric shape. This report derives the complete aligned $a \times b \times c$ volume-filling laws, performs a cubic sweep through $s=243$, identifies all integer interface-boundary closures, and resolves the morphology that emerges as the volume grows.

The central result is exact recursive closure. A $9 \times 9 \times 9$ array of Lucas Supergroups reproduces the complete Lucas-Supergroup cellular census at the Lucas-Supergroup scale. It contains 729 Lucas Supergroups, 19,683 Lucas Groups, and 531,441 Cell3 volumes. The same recursion repeats at side lengths 9, 81, 729, and higher powers of nine in Cell3 units. Between closure scales, the structure passes through exact seven-sheet ternary group scales at factors of three.

The morphology is therefore a fixed anisotropic oblique body with a boundary that grows quadratically, a volume that grows cubically, a bulk fraction that becomes dominant at $s=10$, a laminar sheet count that grows linearly, primitive shell directions that grow quadratically with arithmetic inheritance, and a normalized reciprocal spectrum that converges toward a fixed anisotropic continuum. The cubic $9 \times 9 \times 9$ closure is also the unique isotropic member and minimum-volume member of a finite family of 21 unordered rectangular integer closures.

Central statement: A $9 \times 9 \times 9$ array of Lucas Supergroups is an exact second-order Lucas closure, not merely a larger block.

1. Reader Orientation and Foundational Definitions

The analysis begins at Cell3 and proceeds by exact composition. Cell3 supplies the direct basis B , the reciprocal metric R , the 27-node local scaffold, the six boundary planes, the laminar grade, and the polarity law. A Lucas Group is a $3 \times 3 \times 3$ volume-filling assembly of Cell3 units. A Lucas Supergroup is a $3 \times 3 \times 3$ assembly of Lucas Groups, equivalently a $9 \times 9 \times 9$ assembly of Cell3 volumes.

Cell3 $\rightarrow 3 \times 3 \times 3$ Cell3 = Lucas Group
 Lucas Group $\rightarrow 3 \times 3 \times 3$ Lucas Groups = Lucas Supergroup
 Lucas Supergroup = $9 \times 9 \times 9$ Cell3 volumes = 729 Cell3 volumes

Larger structures are formed by placing complete Lucas Supergroups face-to-face along the three primitive channels. The assembly is described by three positive integers a , b , and c . These integers count Lucas Supergroups along the three basis directions. Because the basis is oblique, the address array is cubical in primitive coordinates but is a parallelepiped in ICRS physical coordinates.

Two resolutions are used throughout. Coarse resolution treats one Lucas Supergroup as one cubical volume. Fine resolution resolves every underlying Cell3 volume. Closure, boundary, interface, sheet, and spectrum statements must therefore be attached to the resolution at which they are evaluated. This level-relative discipline is essential to the recursive structure.

2. Locked Cell3 Geometry

2.1 Authoritative direct and reciprocal bases

The Cell3 direct basis is the 3×3 matrix whose columns are the primitive edge vectors. Coordinates are expressed in ICRS Cartesian nanometers. The direct metric is $M = B^T B$ and the reciprocal quadratic metric is $R = M^{-1}$.

Matrix	Row 1	Row 2	Row 3
B (nm)	0.327501242055891	0.0353307758438284	-0.0101293176267288
	-0.0548970078840064	0.207319850440192	-0.38978770236626
	-0.0374875028947299	-0.547401085388508	-0.689734408367268
M (nm ²)	0.11167605789605	0.0207103332806058	0.0439372350962198
	0.0207103332806058	0.343877732392787	0.296392758961523
	0.0439372350962198	0.296392758961523	0.627770610077296
R (nm ⁻²)	9.20802552945755	0.00153480242819255	-0.645188035699939
	0.00153480242819259	4.90339831236799	-2.31517558441639
	-0.645188035699939	-2.31517558441639	2.73117127471169

Quantity	a channel	b channel	c channel
Edge length (nm)	0.334179679059	0.586410890411	0.792319765043
Opposite-face spacing (nm)	0.329546529695	0.451597379721	0.605097741519
Associated face area (nm ²)	0.357809539129	0.261106235790	0.194869165460

The Cell3 volume is 0.117914891911745 nm³ and the body diagonal is 1.343653621676 nm. The included angles are $\alpha=83.933483012$ degrees, $\beta=80.448130018$ degrees, and $\gamma=50.363231814$ degrees. These non-right angles are preserved at every scale.

2.2 The complete 27-node Cell3 scaffold

Each local address $u=(u_1, u_2, u_3)$ uses u_i in $\{-1, 0, 1\}$. The physical position is $r(u)=B u / 2$. The number of zero address coordinates determines the cellular class: no zeros gives a corner, one zero gives an edge center, two zeros gives a face center, and three zeros gives the body center. The laminar grade is $\lambda=u_1+u_2+u_3$ and the polarity is $\chi=(-1)^\lambda$. Antipodal addresses u and $-u$ occupy opposite positions and have equal polarity.

Locked Cell3: all 27 nodes and exact oblique boundary

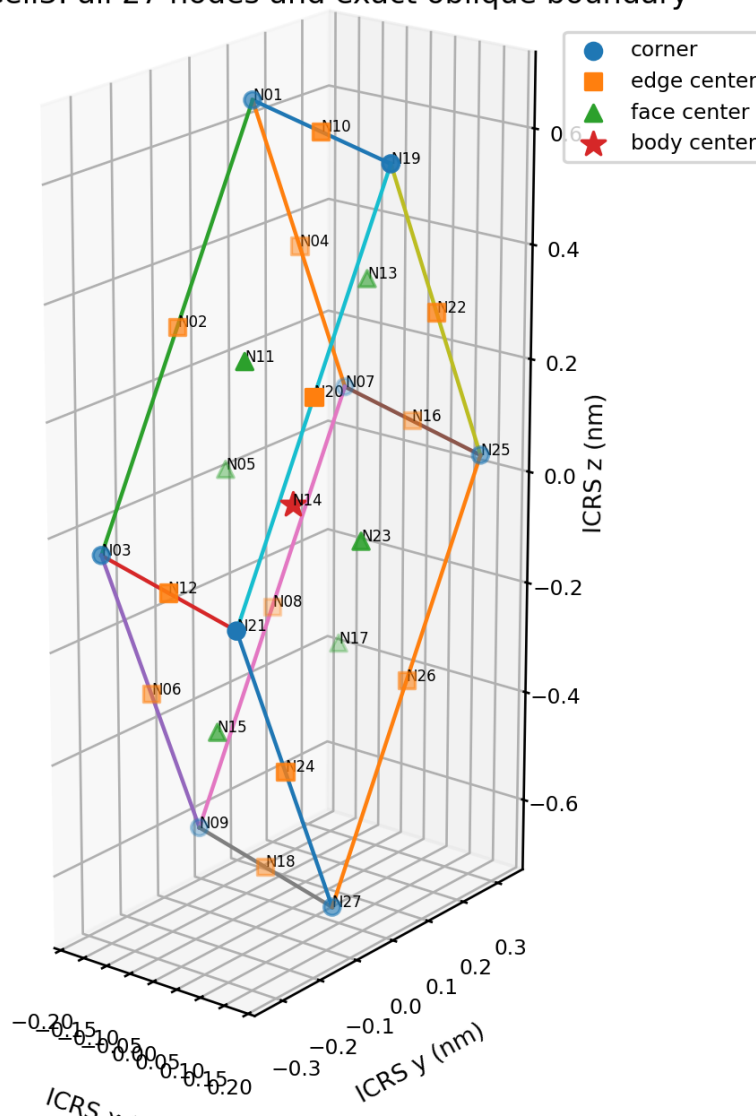


Figure 2.1. The locked Cell3 with all 27 nodes. Node labels N01 through N27 map to the exact address and coordinate ledger in Appendix B. The plotted boundary is the parallelepiped generated by the authoritative basis B.

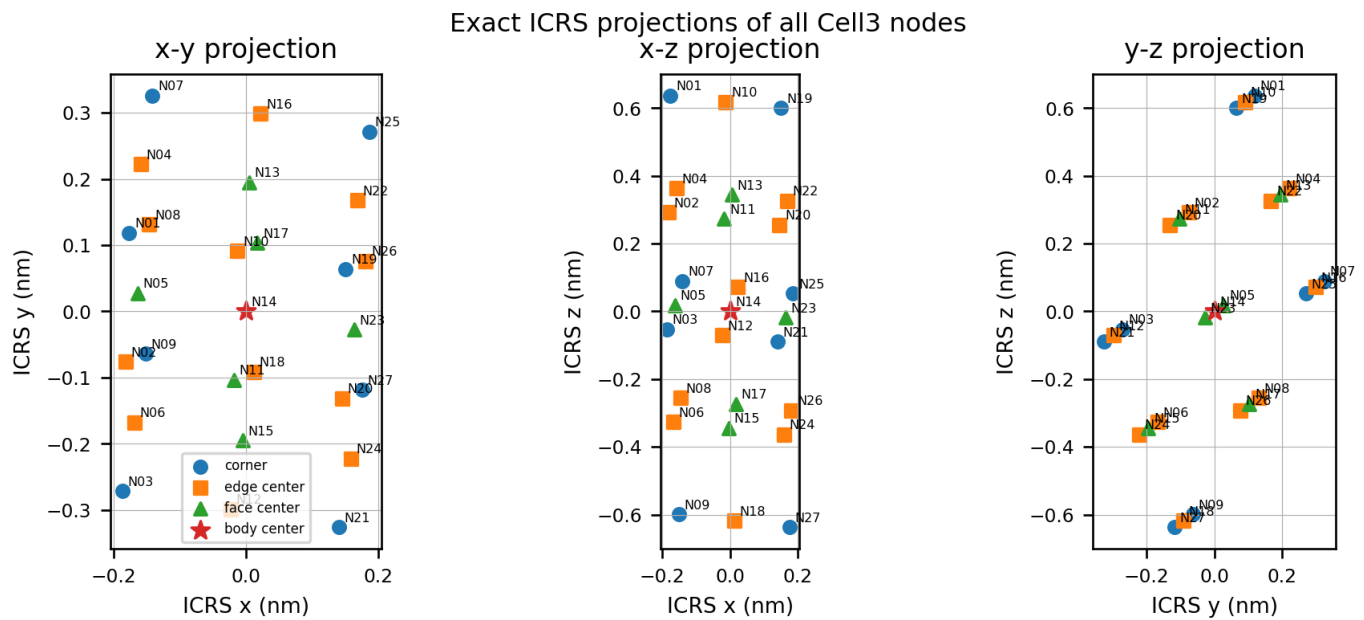


Figure 2.2. Orthographic ICRS projections of all 27 Cell3 nodes. The projections use physical nanometer coordinates rather than schematic primitive-coordinate spacing.

The node census contains 8 corners, 12 edge centers, 6 face centers, and 1 body center. The polarity census contains 13 positive nodes and 14 negative nodes. The seven lamina grades -3 through +3 have populations 1, 3, 6, 7, 6, 3, 1.

2.3 Six exact boundary planes

Cell3 has three opposite plane families. The a channel is bounded by the two bc faces, the b channel by the two ac faces, and the c channel by the two ab faces. In primitive coordinates the planes satisfy $e_i \wedge T B^{-1} r = \pm 1/2$. Every face contains four corners, four edge centers, and one face center.

Six exact Cell3 boundary planes with all face nodes

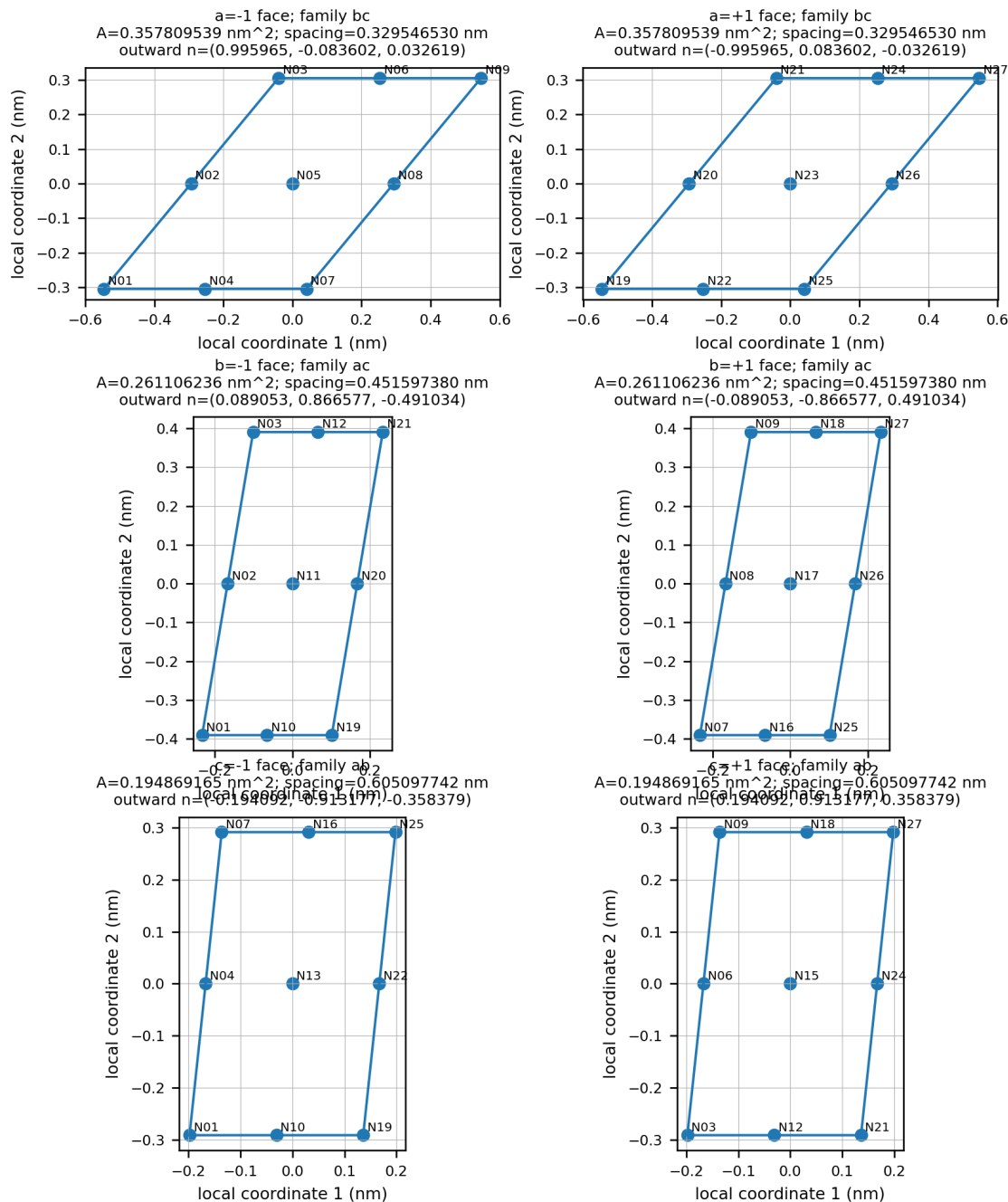


Figure 2.3. All six exact Cell3 boundary planes. Each panel is expressed in an orthonormal basis lying within the physical plane, so distances and areas are dimensionally correct. Every face node is labeled with its Cell3 node identifier, and each panel reports the exact area, opposite-face spacing, and outward ICRS unit normal.

3. The Fully Defined Lucas Supergroup

3.1 Exact construction

Nine Cell3 intervals are placed along each primitive axis. The outer basis is 9B. The resulting volume contains $9^3=729$ Cell3 volumes. These volumes also partition into $3 \times 3 \times 3=27$ Lucas Groups, with each Lucas Group containing 27 Cell3 volumes.

```

outer basis = 9 B
Lucas Groups = 27
Cell3 volumes = 729

```

The outer edge lengths are 3.007617111532 nm, 5.277698013700 nm, and 7.130877885384 nm. The opposite-face spans are 2.965918767259 nm, 4.064376417492 nm, and 5.445879673667 nm. The volume is 85.959956203662 nm³ and the body diagonal is 12.092882595083 nm.

3.2 Complete cellular census

For a cubical complex with m intervals per axis, the generating polynomial is $F_m(t) = ((m+1) + m t)^3$. Setting $m=9$ produces the complete Lucas-Supergroup f -vector.

```

F_9(t) = (10 + 9 t)^3
        = 1000 + 2700 t + 2430 t^2 + 729 t^3

```

Object	Count
Corner vertices	1,000
Edges	2,700
Faces	2,430
Cell3 volumes	729
Complete barycentric states	6,859
Refined laminar sheets	55
Central sheet population	271
Internal Cell3 interfaces	1,944
Boundary barycentric states	1,946
Cell-center graph cycle rank	1,216
Primitive antipodal outer-shell directions	864
Emily $n=19$ nullity	55
Emily $n=19$ rank	6,804

The identity $1946=1944+2$ is the interface-boundary closure that uniquely selects $m=9$ among positive cubic scales. This identity becomes the recursive criterion when complete Lucas Supergroups are treated as the coarse cells of a larger volume.

4. General Aligned Volume-Filling Law

4.1 The $a \times b \times c$ assembly

Let a , b , and c be the numbers of complete Lucas Supergroups along the three primitive basis channels. The outer basis is obtained by scaling the three columns independently.

```

B_out = 9 B diag(a,b,c)

```

The assembly contains abc Lucas Supergroups, $27abc$ Lucas Groups, and $729abc$ Cell3 volumes. The coarse cellular generating polynomial separates by axis.

```

F_(a,b,c)(t) = [(a+1)+a t][(b+1)+b t][(c+1)+c t]

f0 = (a+1)(b+1)(c+1)
f1 = a(b+1)(c+1) + (a+1)b(c+1) + (a+1)(b+1)c
f2 = ab(c+1) + a(b+1)c + (a+1)bc
f3 = abc

```

The alternating sum $f_0 - f_1 + f_2 - f_3$ is one for every positive a , b , and c . Thus every filled rectangular assembly remains a contractible three-dimensional body with Euler characteristic one. Its boundary remains a topological two-sphere with Euler characteristic two.

4.2 Coarse refined-state, boundary, interface, and cycle laws

$$\begin{aligned} Q_SG &= (2a+1)(2b+1)(2c+1) \\ B_SG &= 8(ab+ac+bc) + 2 \\ I_SG &= 3abc - (ab+ac+bc) \\ E_ext &= 2(ab+ac+bc) \\ \text{beta_1} &= 2abc - (ab+ac+bc) + 1 \end{aligned}$$

Q_SG counts all coarse barycentric states: vertices, edge centers, face centers, and volume centers. B_SG counts those states on the outer boundary. I_SG counts shared faces between adjacent Lucas Supergroups. E_ext counts exposed Lucas-Supergroup faces. The cycle rank beta_1 belongs to the nearest-neighbor graph of Lucas-Supergroup centers.

4.3 Fine Cell3 resolution

Each coarse interval contains nine Cell3 intervals. Replacing a,b,c by 9a,9b,9c gives the complete fine-resolution laws.

$$\begin{aligned} Q_Cell3 &= (18a+1)(18b+1)(18c+1) \\ B_Cell3 &= 648(ab+ac+bc) + 2 \\ I_Cell3 &= 2187abc - 81(ab+ac+bc) \\ E_Cell3,ext &= 162(ab+ac+bc) \end{aligned}$$

These formulas are exhaustive for every aligned rectangular volume-filling set. No numerical sweep is required to extend them beyond the tabulated range because the counts are exact polynomials in a, b, and c.

4.4 Physical dimensions and area

$$\begin{aligned} V &= V_SG \ abc \\ S &= 2[A_ab \ ab + A_ac \ ac + A_bc \ bc] \end{aligned}$$

Here $V_SG=85.959956203662 \text{ nm}^3$, $A_ab=15.784402402299 \text{ nm}^2$, $A_ac=21.149605099007 \text{ nm}^2$, and $A_bc=28.982572669413 \text{ nm}^2$. The physical surface is metric-sensitive: interchanging a, b, and c changes the physical area because the three face families have different areas.

5. Compact Cubic Growth

5.1 Exact cubic laws

The compact self-similar family uses $a=b=c=s$. It contains s^3 Lucas Supergroups and $729s^3$ Cell3 volumes.

$$\begin{aligned} N_SG &= s^3 \\ I_SG &= 3s^2(s-1) \\ E_ext &= 6s^2 \\ \text{beta_1} &= 2s^3 - 3s^2 + 1 \\ Q_SG &= (2s+1)^3 \\ B_SG &= 24s^2 + 2 \\ N_Cell3 &= 729s^3 \\ Q_Cell3 &= (18s+1)^3 \\ B_Cell3 &= 1944s^2 + 2 \\ I_Cell3 &= 2187s^3 - 243s^2 \\ E_Cell3,ext &= 486s^2 \end{aligned}$$

The exposed-face density and boundary-state fraction decrease with inverse linear scale.

$$\begin{aligned} E_Cell3,ext / N_Cell3 &= 2/(3s) \\ B_Cell3 / Q_Cell3 &\text{ approaches } 1/(3s) \text{ for large } s \end{aligned}$$

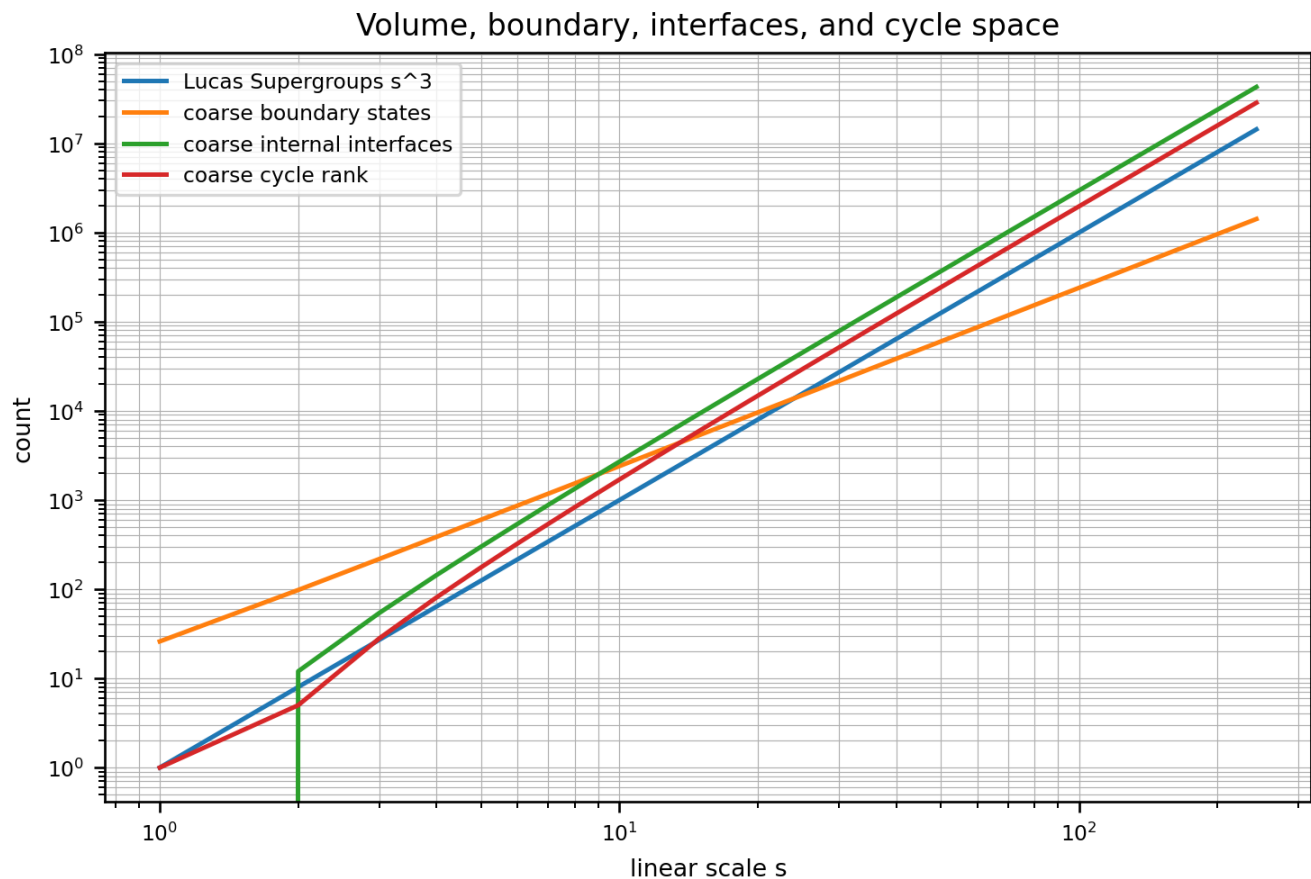


Figure 5.1. Exact polynomial scaling through $s=243$. Volume and cycle-space counts are cubic at leading order, while boundary states and exposed faces are quadratic.

5.2 Local coordination morphology

The coarse center lattice has four local environments. Corners have three neighbors, edge-interior sites have four, face-interior sites have five, and bulk sites have six.

Environment	Coordination	Population for $s \geq 2$
Corner	3	8
Edge interior	4	$12(s-2)$
Face interior	5	$6(s-2)^2$
Bulk	6	$(s-2)^3$

mean coordination = $6(1 - 1/s)$
 bulk fraction = $[(s-2)/s]^3$

A fully coordinated bulk Lucas Supergroup first appears at $s=3$. The bulk becomes the majority at $s=10$ because $(8/10)^3=0.512$. The mean coordination tends to six as the influence of the outer boundary becomes negligible.

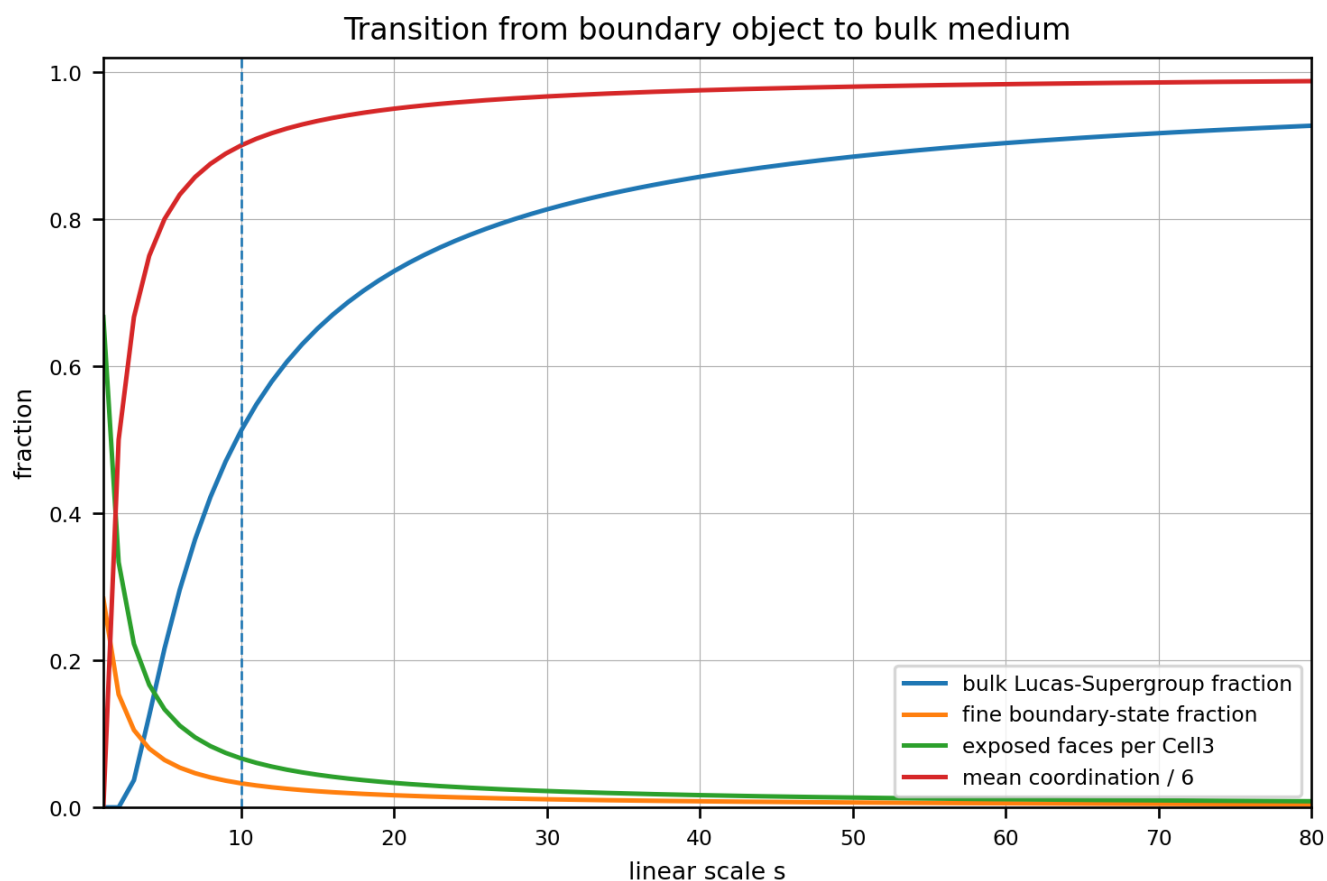


Figure 5.2. Boundary-to-bulk transition. The bulk fraction exceeds one-half at $s=10$, while both fine boundary-state fraction and exposed-face density decrease toward zero.

5.3 Dimensionally exact growth morphology

Exact aligned cubic volume-filling growth in the locked oblique basis

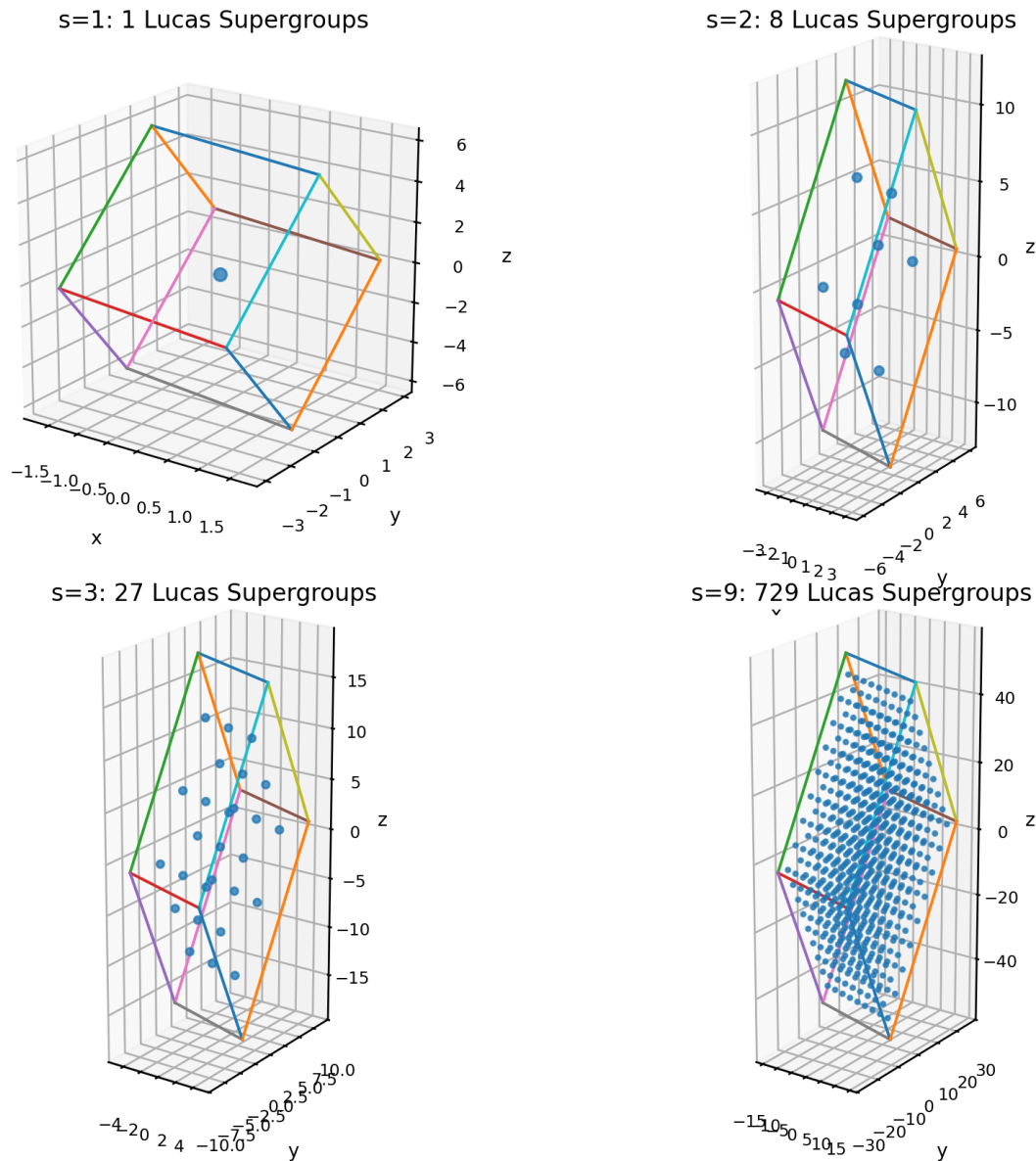


Figure 5.3. Exact Lucas-Supergroup center arrays and outer oblique bodies for $s=1,2,3,9$. Equal primitive counts preserve the oblique physical morphology at every scale.

s	Lucas SG	Cell3	Bulk SG	Bulk frac.	Cycle rank	Fine sheets	Primitive dirs.
1	1	729	0	0.000000	0	55	864
2	8	5,832	0	0.000000	5	109	2,592
3	27	19,683	1	0.037037	28	163	7,776
8	512	373,248	216	0.421875	833	433	41,472
9	729	531,441	343	0.470508	1,216	487	69,984
10	1,000	729,000	512	0.512000	1,701	541	62,208
27	19,683	14,348,907	15,625	0.793832	37,180	1,459	629,856
81	531,441	387,420,489	493,039	0.927740	1,043,200	4,375	5,668,704
243	14,348,907	10,460,353,203	13,997,521	0.975511	28,520,668	13,123	51,018,336

6. Exact Morphological Transitions

6.1 s=2: bridge and loop formation

Eight Lucas Supergroups form the first filled three-dimensional block. All eight remain boundary objects, but twelve inter-supergroup contacts appear. The center graph has cycle rank $12-8+1=5$. The structure is therefore no longer a set of isolated closures; it possesses collective closed paths.

6.2 s=3: reappearance of the seven-sheet polarity complex

A $3 \times 3 \times 3$ array contains 27 Lucas Supergroups. Center addresses again use $\{-1,0,1\}^3$. The seven laminar grades -3 through +3 reappear with populations 1,3,6,7,6,3,1. The central Lucas Supergroup is the first six-neighbor bulk object. The seven-sheet Cell3 polarity morphology has therefore returned one full hierarchy level above the original Cell3 scaffold.

6.3 s=8: first additional coarse adjacency resonance

The open cubic nearest-neighbor spectrum has eigenvalues $2[\cos(p\pi/(s+1)) + \cos(q\pi/(s+1)) + \cos(r\pi/(s+1))]$. Odd s always contains the complement-center zero-mode family. At $s=8$ the first additional even-scale 120-degree cosine cancellation becomes available. The exact coarse sweep gives 12 zero modes. This marks the first resonance produced by the denominator $s+1=9$ rather than by central-index complementarity.

6.4 s=9: second-order closure

$$B_s - I_s - 2 = 3s^2(9-s)$$

The residual is positive below nine, zero at nine, and negative above nine. Thus $s=9$ is the unique isotropic coarse scale satisfying $B_s=I_s+2$. The $9 \times 9 \times 9$ array reproduces the complete Lucas-Supergroup census at the Lucas-Supergroup scale.

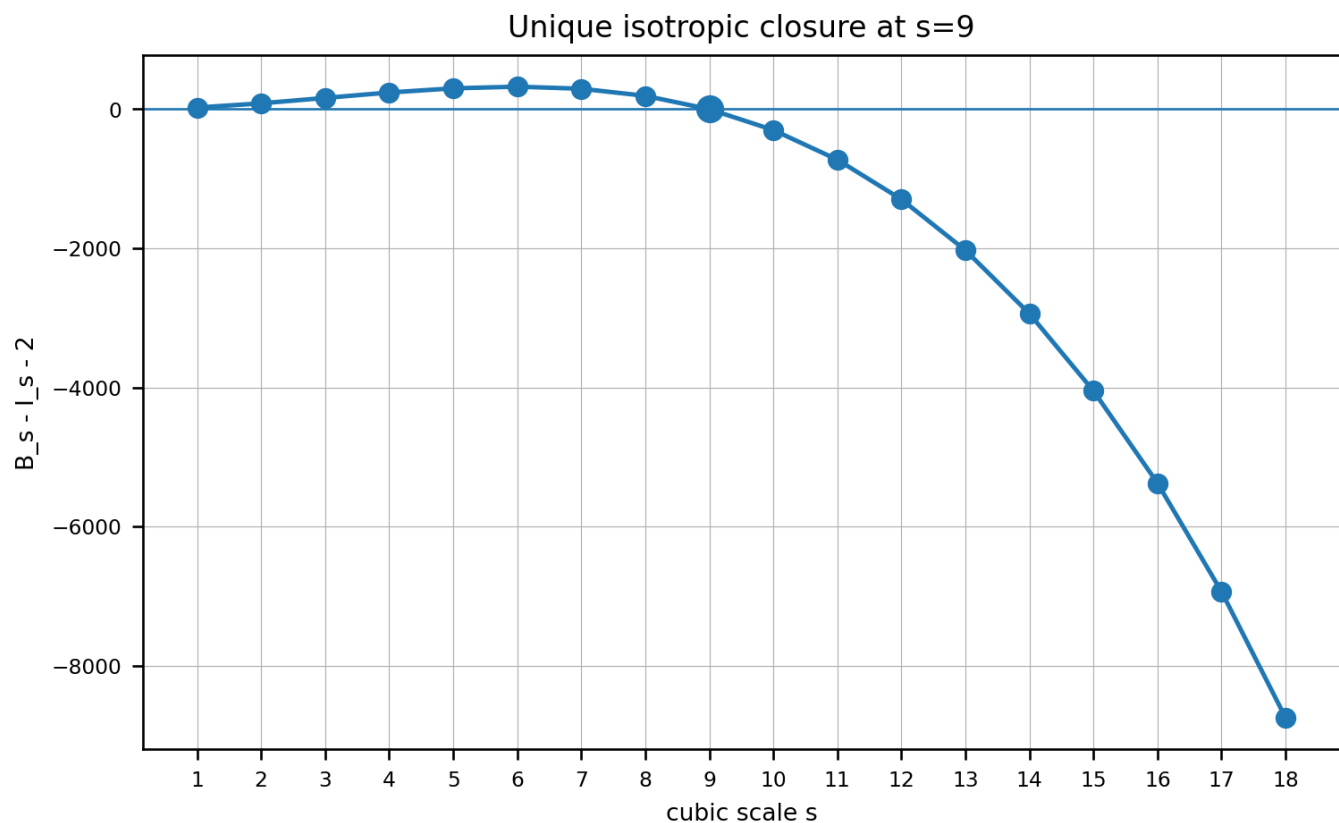


Figure 6.1. The cubic interface-boundary residual has a unique positive zero at $s=9$. This is the exact second-order closure.

Second-order quantity	Count
Lucas Supergroups	729

Second-order quantity	Count
Lucas Groups	19,683
Cell3 volumes	531,441
Coarse f-vector	(1000, 2700, 2430, 729)
Coarse refined states	6,859
Coarse internal interfaces	1,944
Coarse boundary states	1,946
Coarse cycle rank	1,216
Fine f-vector	(551368, 1633932, 1614006, 531441)
Fine refined states	4,330,747
Fine internal interfaces	1,574,640
Fine boundary states	157,466
Fine exposed faces	39,366

Closure is level-relative. At coarse Lucas-Supergroup resolution, the 81B Cell3 body has the same $1946=1944+2$ closure as the original 9B body. At fine Cell3 resolution it is a side-81 block and therefore has its own much larger boundary and interface counts.

6.5 s=10: bulk dominance

At $s=10$, 512 of 1,000 Lucas Supergroups are six-neighbor bulk objects. This is the first scale at which bulk morphology exceeds boundary morphology.

6.6 s=27: seven-sheet assembly of second-order closures

A side-27 array partitions into $3 \times 3 \times 3=27$ second-order closure blocks, each of side nine in Lucas-Supergroup units. The quotient array of closure blocks again has the seven-sheet populations 1,3,6,7,6,3,1.

6.7 s=81: third-order closure

A side-81 array partitions into $9 \times 9 \times 9=729$ second-order closures. Their quotient complex is another exact Lucas closure. The full body contains 531,441 Lucas Supergroups and 387,420,489 Cell3 volumes.

7. Complete Rectangular Closure Family

7.1 Closure equation

The cubic closure is part of a finite rectangular family. Applying $B_SG=I_SG+2$ to an $a \times b \times c$ assembly gives

$$\begin{aligned} 8(ab+ac+bc)+2 &= 3abc - (ab+ac+bc)+2 \\ abc &= 3(ab+ac+bc) \\ 1/a + 1/b + 1/c &= 1/3 \end{aligned}$$

For ordered positive integers $a \leq b \leq c$, exactly 21 solutions exist. Permuting axis assignments produces 106 oriented closures because the three physical basis channels are distinct.

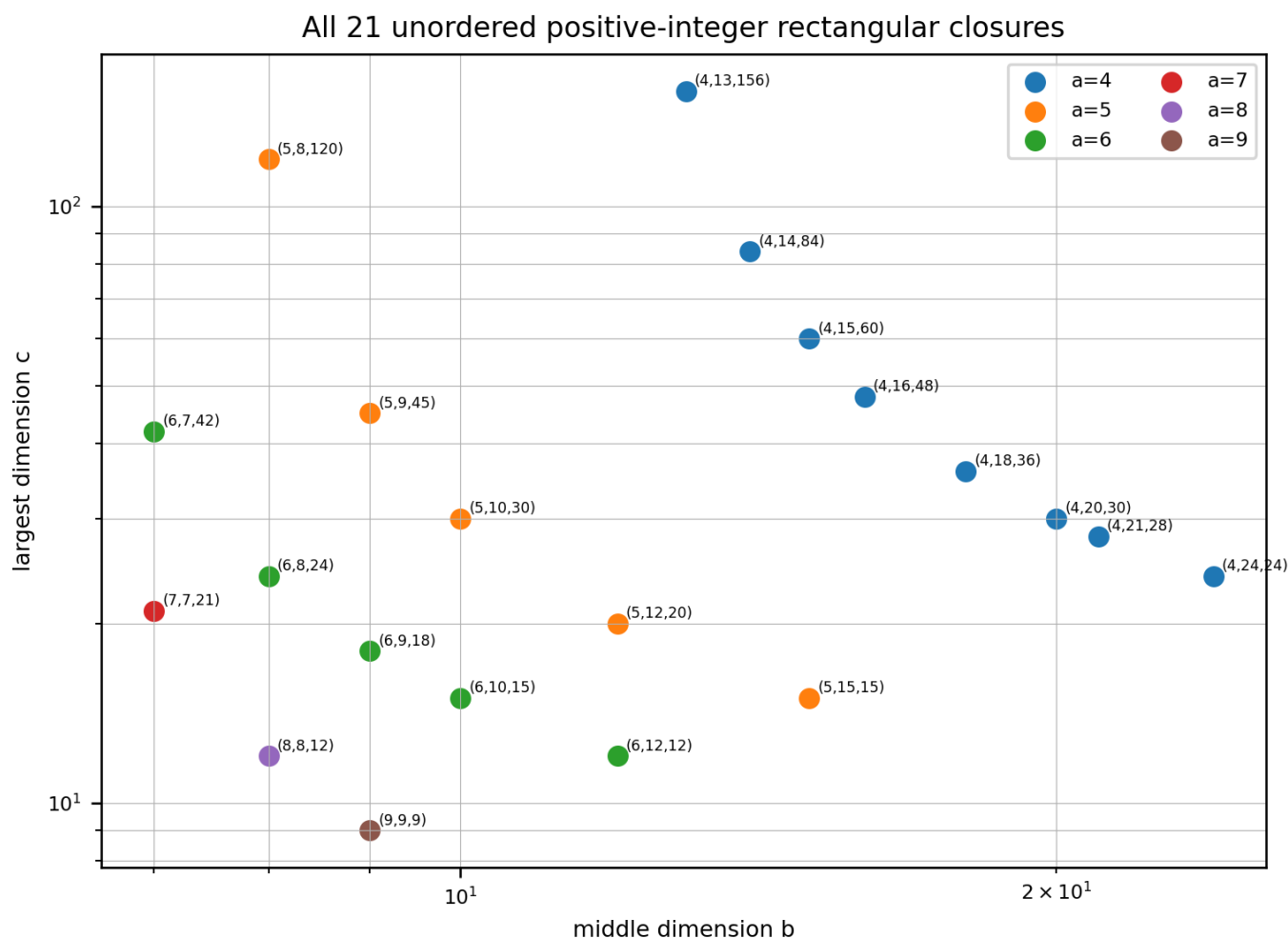


Figure 7.1. The complete set of 21 unordered positive-integer closure triples. Logarithmic axes reveal the compact isotropic solution and the elongated anisotropic families.

a	Allowed (b,c) pairs
4	(13,156), (14,84), (15,60), (16,48), (18,36), (20,30), (21,28), (24,24)
5	(8,120), (9,45), (10,30), (12,20), (15,15)
6	(7,42), (8,24), (9,18), (10,15), (12,12)
7	(7,21)
8	(8,12)
9	(9,9)

7.2 Universal properties of every rectangular closure

Let $N=abc$. The closure equation implies $ab+ac+bc=N/3$. Substitution produces universal closure laws independent of aspect ratio.

$$\begin{aligned}
 I_{SG} &= 8N/3 \\
 B_{SG} &= 8N/3 + 2 \\
 E_{ext} &= 2N/3 \\
 \text{beta}_1 &= 5N/3 + 1 \\
 \text{mean coordination} &= 16/3
 \end{aligned}$$

Every exact rectangular closure therefore has the same mean coordination, exposed-face density, and cycle-rank density. Aspect ratio changes physical dimensions and metric surface area, but not these normalized combinatorial closure properties.

The $9 \times 9 \times 9$ solution is the unique isotropic closure. It is also the smallest closure by unit count: 729 Lucas Supergroups, compared with 768 for the next solution $8 \times 8 \times 12$.

8. Physical Morphology and Metric-Sensitive Orientation

8.1 Fixed anisotropic shape under equal-count growth

For cubic growth the physical outer body remains an oblique parallelepiped. Every edge, face span, and body diagonal scales linearly with s , every face area scales as s^2 , and volume scales as s^3 . Angles, face normals, principal directions, and reciprocal condition number remain unchanged.

```
edge ratio = 1 : 1.754777226617 : 2.370939391867
face-span ratio = 1 : 1.370359991770 : 1.836152673426
reciprocal condition number = 7.593830822301
```

Scale growth therefore creates a larger anisotropic medium rather than a body that tends toward a sphere. The normalized direct and reciprocal geometries remain exactly fixed.

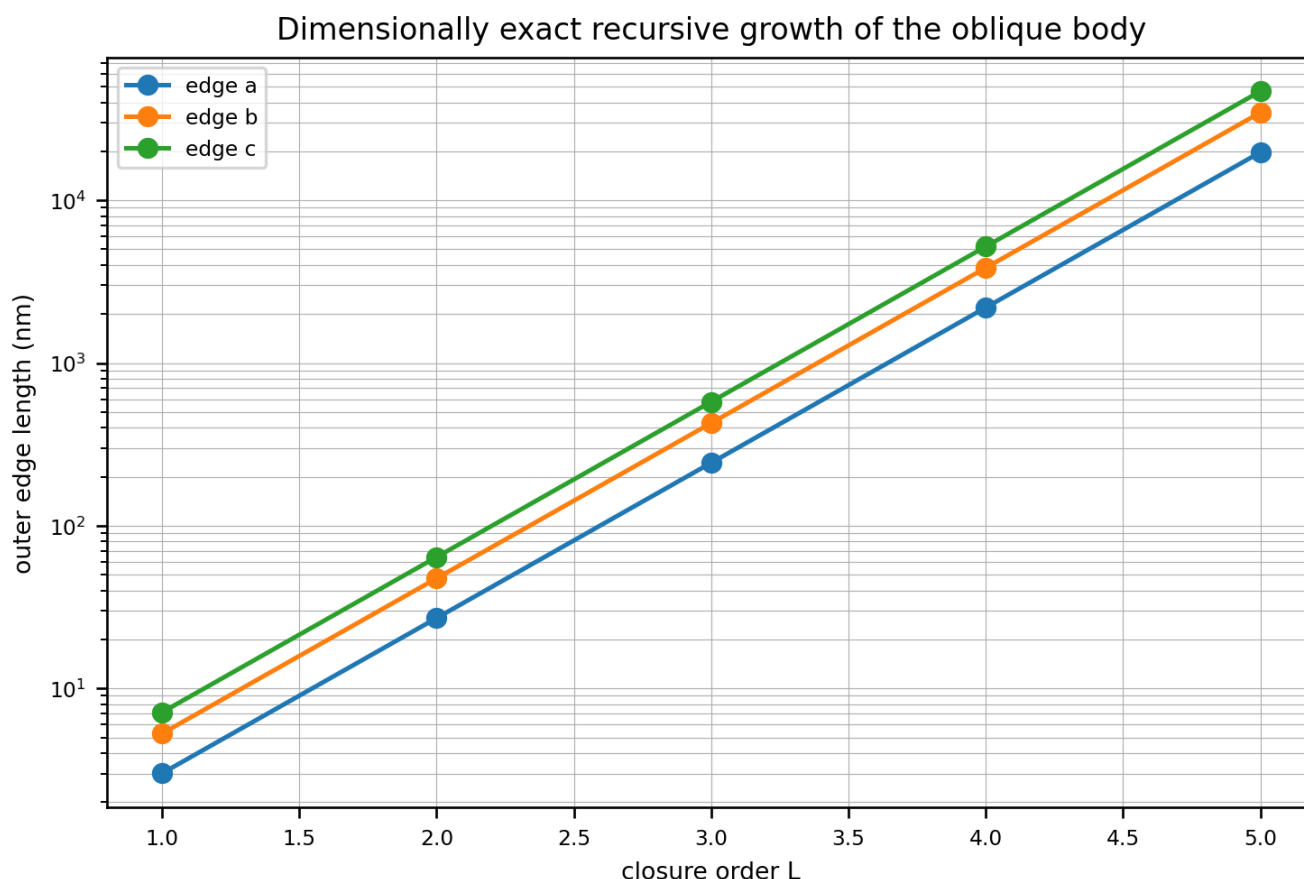


Figure 8.1. Exact recursive edge growth. All three edges increase by a factor of nine at each closure order while retaining their fixed ratios.

8.2 Cubic surface-to-volume law

```
S(s) = 131.833160341439 s^2 nm^2
V(s) = 85.959956203662 s^3 nm^3
S/V = 1.533657835156/s nm^-1
```

The physical influence of the outer surface decreases as $1/s$. This is the geometric counterpart of the combinatorial boundary-to-volume transition.

8.3 Continuous minimum-area count ratio

For fixed abc , minimizing physical surface area gives $a:b:c = A_{bc}:A_{ac}:A_{ab}$. Numerically, the minimum-area count ratio is

```
a:b:c = 1.836152673426 : 1.339905341993 : 1
```

This differs from equal-count compactness because physical faces have unequal areas. Equal a,b,c minimizes the number of exposed coarse faces for fixed unit count, while the metric-weighted ratio minimizes actual nanometer-squared area.

8.4 The 106 oriented exact closures

Across all 106 oriented integer closures, physical S/V ranges from $0.133987481545 \text{ nm}^{-1}$ for orientation (156, 13, 4) to $0.208788331923 \text{ nm}^{-1}$ for orientation (4, 13, 156). The two extremes use the same unordered closure triple but assign its long and short counts to different physical basis channels.

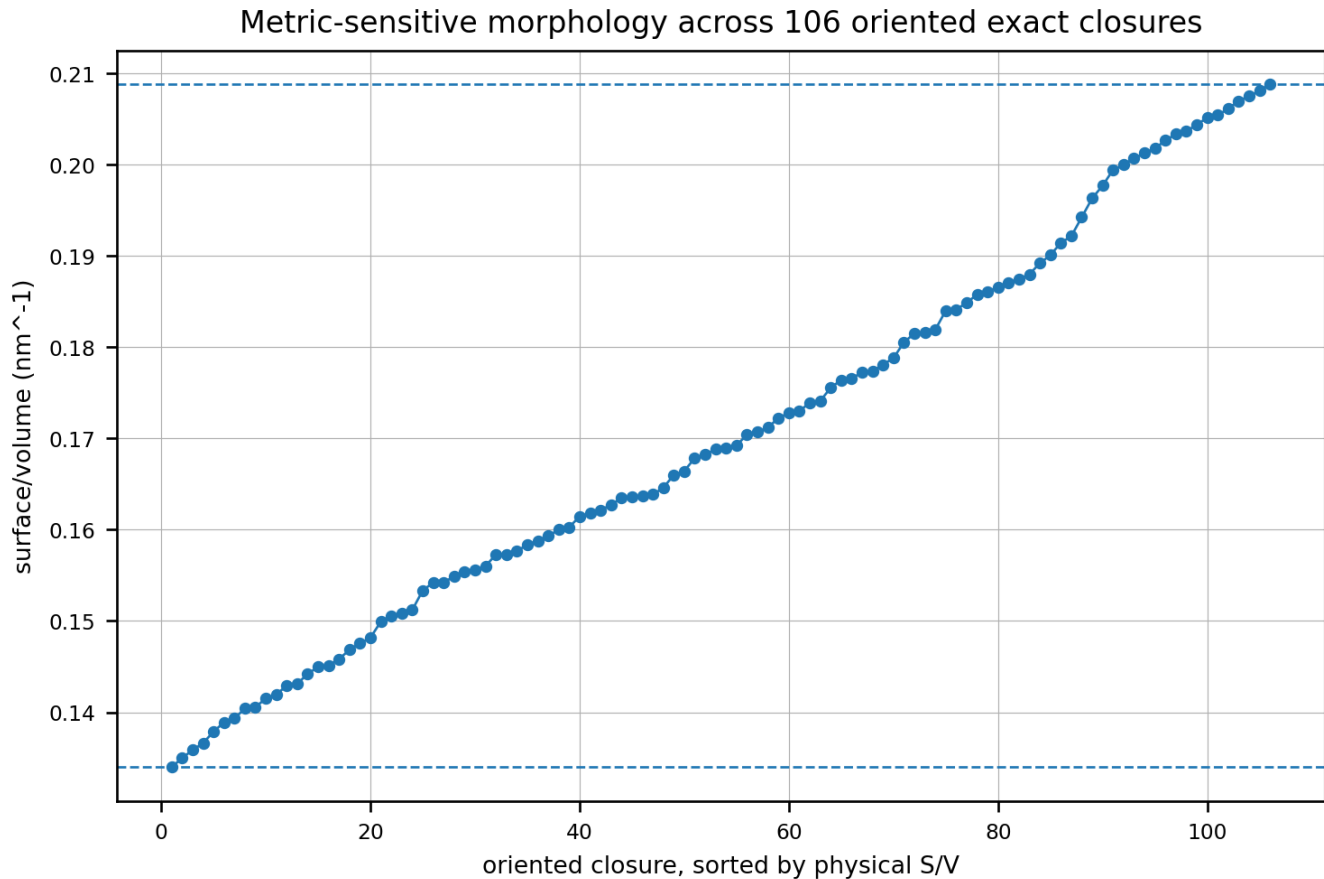


Figure 8.2. Physical surface-to-volume ratios for all 106 oriented exact closures, sorted from minimum to maximum. Axis assignment is a physical degree of morphology because the three face families have unequal areas.

8.5 Exact second-order physical dimensions

The second-order closure has outer basis 81B. Its outer edge lengths are 27.068554003858 nm, 47.499282123302 nm, and 64.177901013958 nm. Its opposite-face spans are 26.693268905331 nm, 36.579387781424 nm, and 49.012917061003 nm. Its body diagonal is 108.835943355750 nm and its volume is $62,664.808072469 \text{ nm}^3$.

9. Recursive Interface Structure

9.1 Local 3:1:1 decomposition

Within one Lucas Supergroup, the 1,944 internal Cell3 interfaces divide into 1,458 interfaces internal to Lucas Groups and 486 interfaces between Lucas Groups. The outer boundary contains 486 exposed Cell3 faces. The exact local ratio is therefore 3:1:1.

$$\text{intragroup} : \text{intergroup} : \text{external} = 1458 : 486 : 486 = 3 : 1 : 1$$

9.2 Closure-order hierarchy

Define closure order L by an outer side of 9^L Cell3 intervals. Order one is the ordinary Lucas Supergroup. Order two is the $9 \times 9 \times 9$ array of Lucas Supergroups.

$$\begin{aligned} B_L &= 9^L B \\ N_{Cell3}(L) &= 729^L \\ N_{SG}(L) &= 729^{(L-1)} \end{aligned}$$

The number of fine Cell3 interfaces contributed at hierarchy level j is

$$C_j = 1944 * 81^{(j-1)} * 729^{(L-j)}, \quad 1 \leq j \leq L$$

Successive coarser levels satisfy $C_{(j+1)}/C_j=1/9$. The full interface count is the finite sum of these hierarchy levels and equals the direct side-length formula.

$$I_L = \sum_j C_j = 3(9^L)^2(9^L-1)$$

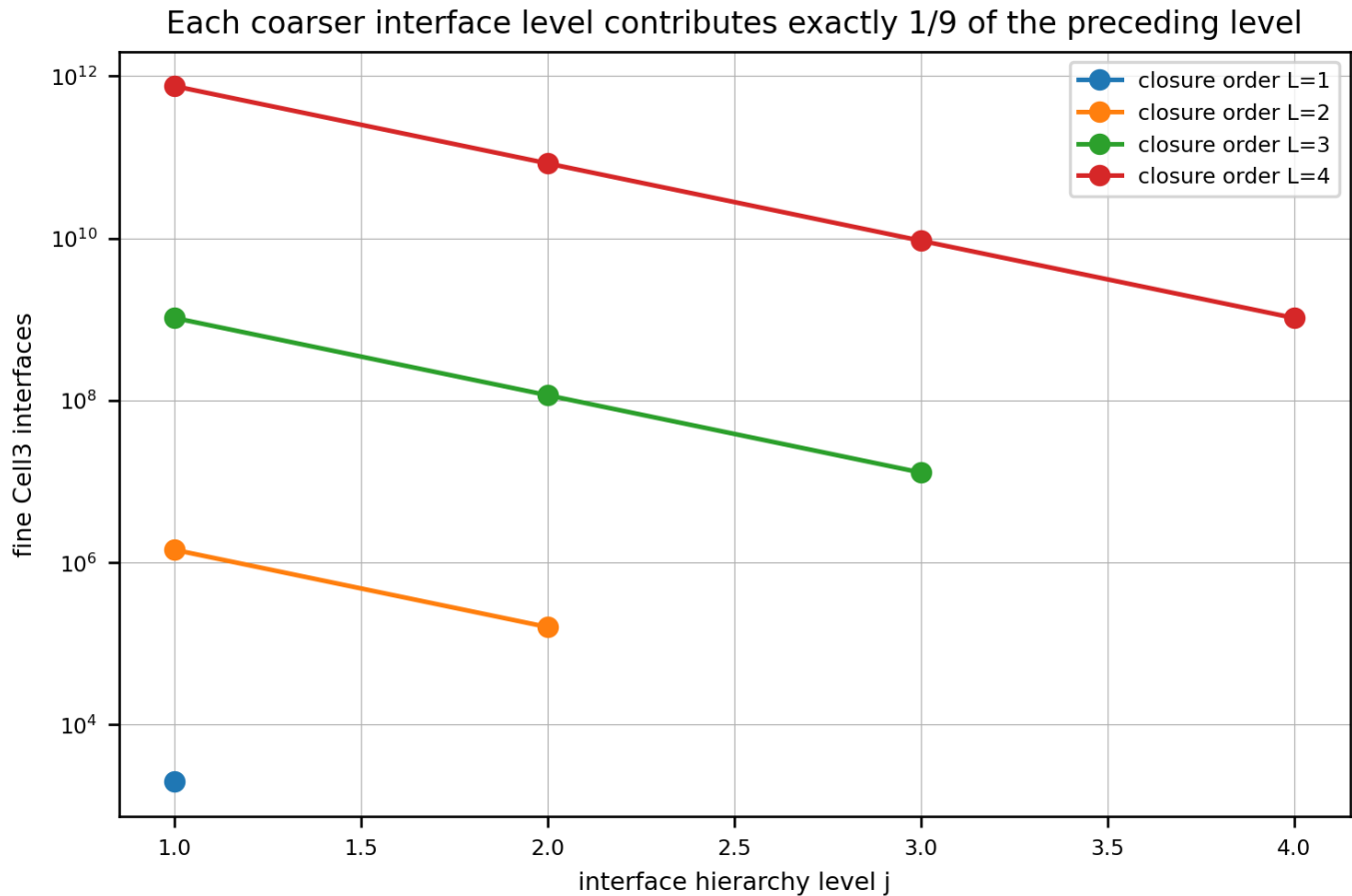


Figure 9.1. Exact hierarchy-level interface contributions for closure orders one through four. Each coarser level is reduced by exactly one-ninth.

9.3 Second-order interface ledger

The 531,441 Cell3 volumes of the second-order closure have 1,574,640 internal Cell3 interfaces. Of these, 1,417,176 lie within the 729 constituent Lucas Supergroups and 157,464 join adjacent Lucas Supergroups. The internal constituent term separates further into 1,062,882 interfaces internal to Lucas Groups and 354,294 interfaces between Lucas Groups. The outer boundary contains 39,366 exposed Cell3 faces.

10. Balanced-Ternary and Polarity Recursion

10.1 Two-digit Lucas-Supergroup address

The Lucas Supergroup address has the exact balanced-ternary decomposition $x=3G+c$, where G and c each range over $\{-1,0,1\}^3$. G identifies the Lucas Group and c identifies the Cell3 position inside that group.

$$\begin{aligned} x &= 3G + c \\ \text{chi}(x) &= \text{chi}(G) \text{chi}(c) \\ \text{lambda}(x) &= 3 \text{lambda}(G) + \text{lambda}(c) \end{aligned}$$

10.2 Arbitrary ternary hierarchy

```
x = sum_(j=0)^(k-1) 3^j c_j
lambda(x) = sum_(j=0)^(k-1) 3^j lambda(c_j)
chi(x) = product_(j=0)^(k-1) chi(c_j)
```

Every power 3^j is odd, so every ternary digit contributes directly to parity. Each digit is therefore a seven-sheet polarity component. A Lucas Supergroup has linear width $9=3^2$ and contains two balanced-ternary digits per axis. Every additional Lucas closure adds two further digits per axis.

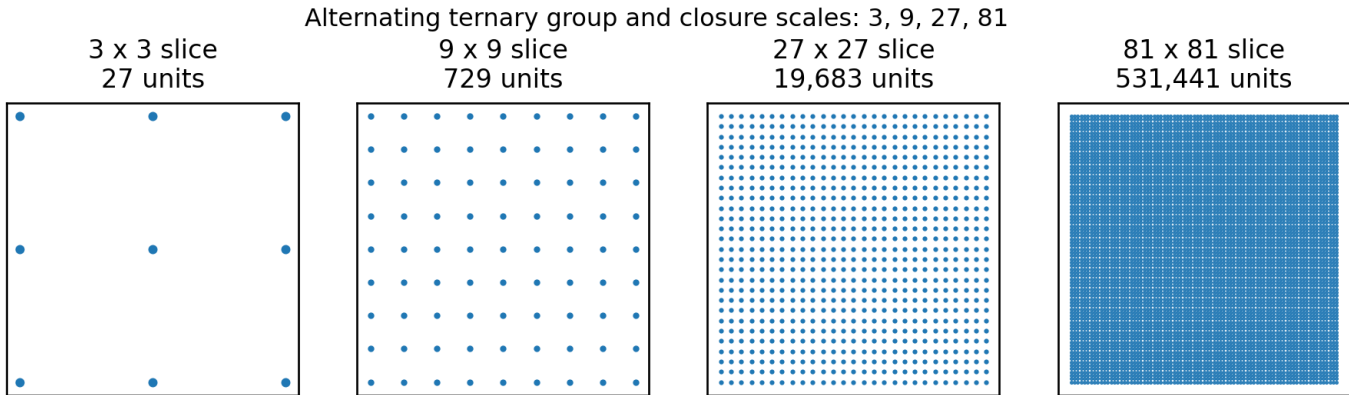


Figure 10.1. Exact center-lattice slices at widths 3, 9, 27, and 81. Widths 3 and 27 are seven-sheet quotient scales; widths 9 and 81 are complete closure scales.

```
3 x 3 x 3: seven-sheet group morphology
9 x 9 x 9: complete Lucas closure
27 x 27 x 27: seven-sheet assembly of closures
81 x 81 x 81: next complete closure
```

11. Laminar-Sheet Growth

11.1 Exact sheet count and population law

For a cubic array of s Lucas Supergroups per axis, the fine refined lattice has primitive radius $R=9s$ and coordinate range $\{-R, \dots, R\}^3$. The laminar grade λ is the sum of the three refined coordinates.

```
number of sheets = 6R+1 = 54s+1
central population = 3R^2+3R+1 = 243s^2+27s+1

P_R(lambda) = 3R^2+3R+1-lambda^2, for |lambda| <= R
P_R(lambda) = C(3R^2-|lambda|, 2), for R <= |lambda| <= 3R
```

The central quadratic core and triangular tails extend the exact radius-nine Lucas-Supergroup sheet law to every larger filled volume. The total population summed across all sheets is $(2R+1)^3$.

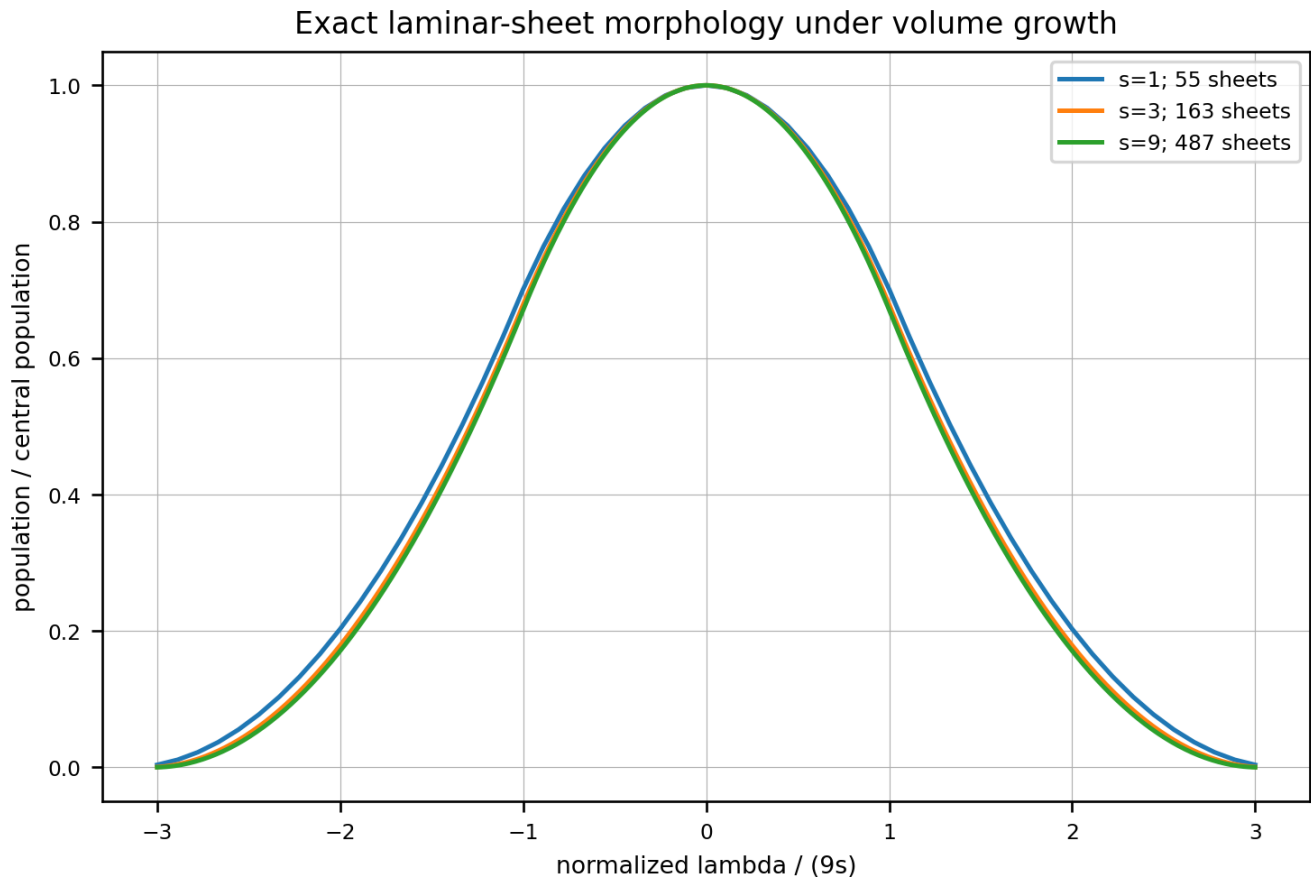


Figure 11.1. Exact normalized laminar profiles for $s=1,3,9$. The profile converges after coordinate and population normalization, while the number of sheets and absolute central population continue to grow.

11.2 Sheet-to-volume scaling

$$P_R(0)/(2R+1)^3 \text{ approaches } 3/(8R)$$

The number of sheets grows linearly and the population of a central sheet grows quadratically, but total volume grows cubically. Every individual sheet therefore becomes a vanishing fraction of a sufficiently large volume, even though its absolute transverse extent continues to increase.

12. Primitive Shell Morphology and Arithmetic Inheritance

12.1 Outer shell counts

The fine outer shell at radius R contains all integer addresses with maximum coordinate magnitude R . Its directed node count and antipodal direction count are

$$\begin{aligned} |S_R| &= 24R^2 + 2 \\ D_{all} &= 12R^2 + 1 \end{aligned}$$

A direction is primitive when the greatest common divisor of its three coordinate magnitudes is one. The primitive antipodal count is governed by the Jordan totient J_2 .

$$\begin{aligned} D_{prim} &= 12 J_2(R) \\ J_2(R) &= R^2 \prod_{p \text{ divides } R} (1 - 1/p^2) \\ D_{inh} &= 12R^2 + 1 - 12J_2(R) \end{aligned}$$

12.2 Pure ternary recursion

When R is a power of three, $J_2(R)=8R^2/9$. Therefore

$$D_{\text{prim}} = (32/3)R^2$$

$$D_{\text{inh}} = (4/3)R^2 + 1$$

For $R=9s$ with s itself a power of three, $D_{\text{prim}}=864s^2$ and $D_{\text{inh}}=108s^2+1$. The inherited shell at radius R is exactly the complete antipodal shell at radius $R/3$.

R	Primitive antipodal	Inherited antipodal
9	864	109
27	7,776	973
81	69,984	8,749
243	629,856	78,733
729	5,668,704	708,589

Arithmetic shell inheritance through $s=81$

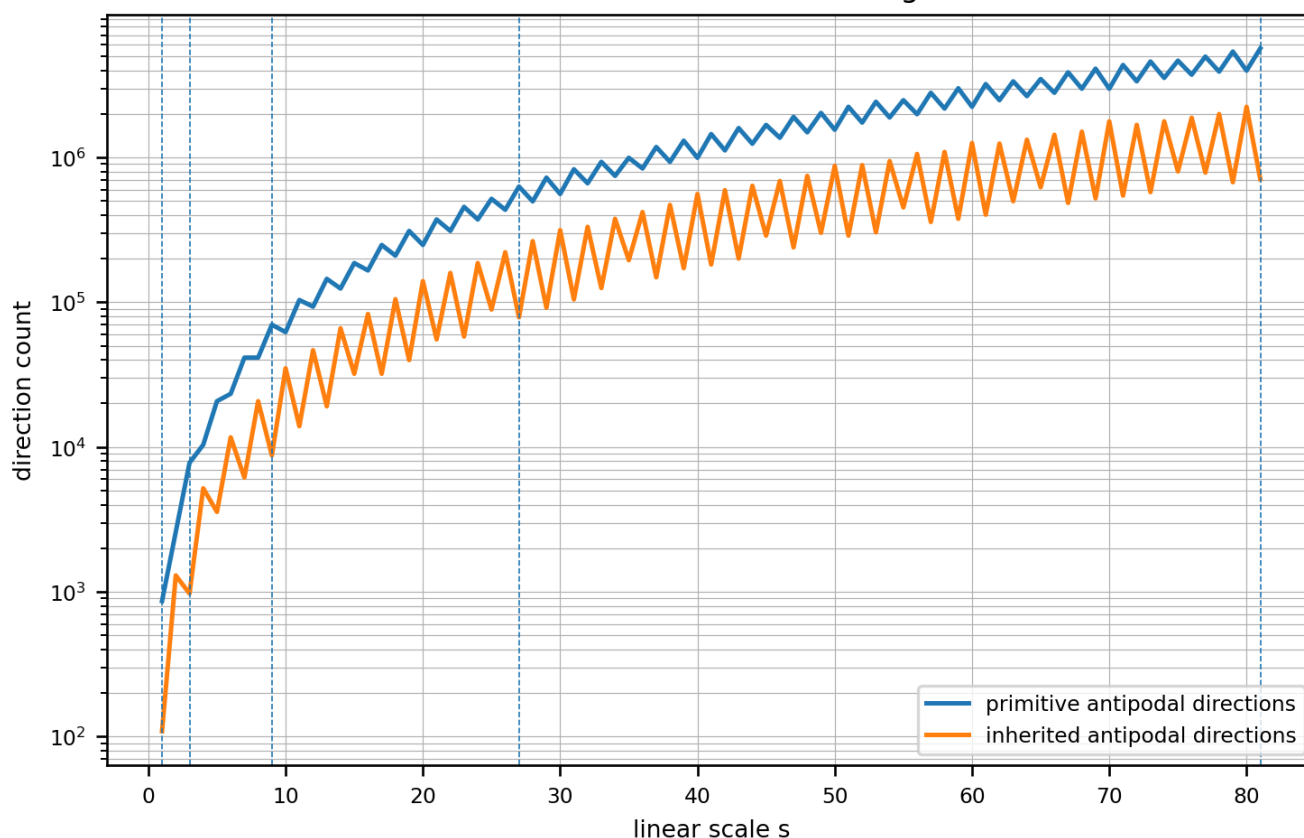


Figure 12.1. Primitive and inherited antipodal direction counts through $s=81$. New prime factors in s create arithmetic changes in the primitive fraction, while pure powers of three follow exact quadratic recursion.

13. Reciprocal q-Spectrum

13.1 Quadratic observation

Every primitive shell address u is assigned the reciprocal scalar $q(u)=u^T R u$. The six independent coefficients of the symmetric metric R define a six-dimensional quadratic observation space. The radius-nine shell has 864 primitive antipodal observations, all with distinct q values at $1e-10$ rounding, while the observation matrix rank remains six.

$$q(u) = u^T R u$$

13.2 Exact mean scaling

Cubic shell symmetry makes the coordinate-square sums equal. For every primitive shell radius $R>1$, the mean coordinate square is $5R^2/9$. Hence

$$\text{mean}(q) = (5/9) R^2 \text{ trace}(R)$$

With $R=9s$ and $\text{trace}(R)=16.842595116537 \text{ nm}^{-2}$, $\text{mean}(q)=757.916780244175 \text{ s}^2 \text{ nm}^{-2}$. At $s=1$ this gives $757.916780244175 \text{ nm}^{-2}$, exactly matching the radius-nine source census.

13.3 Compatibility growth

observation count = 12 J2(9s)
metric rank = 6
compatibility dimension = 12 J2(9s) - 6

The number of primitive observations grows quadratically while the independent metric dimension remains fixed. The compatibility space therefore grows quadratically and records increasingly dense directional consistency with the same six metric coefficients.

13.4 Normalized anisotropic continuum

Dividing q by R^2 removes the trivial scale factor. The primitive distributions at $R=9,27,81$ densify toward a fixed anisotropic continuum. The exact normalized mean is $(5/9)\text{trace}(R)=9.356997286965 \text{ nm}^{-2}$. The continuum Rayleigh limits over the cube shell are fixed by the reciprocal metric and remain anisotropic.

Convergence of the normalized primitive q-spectrum

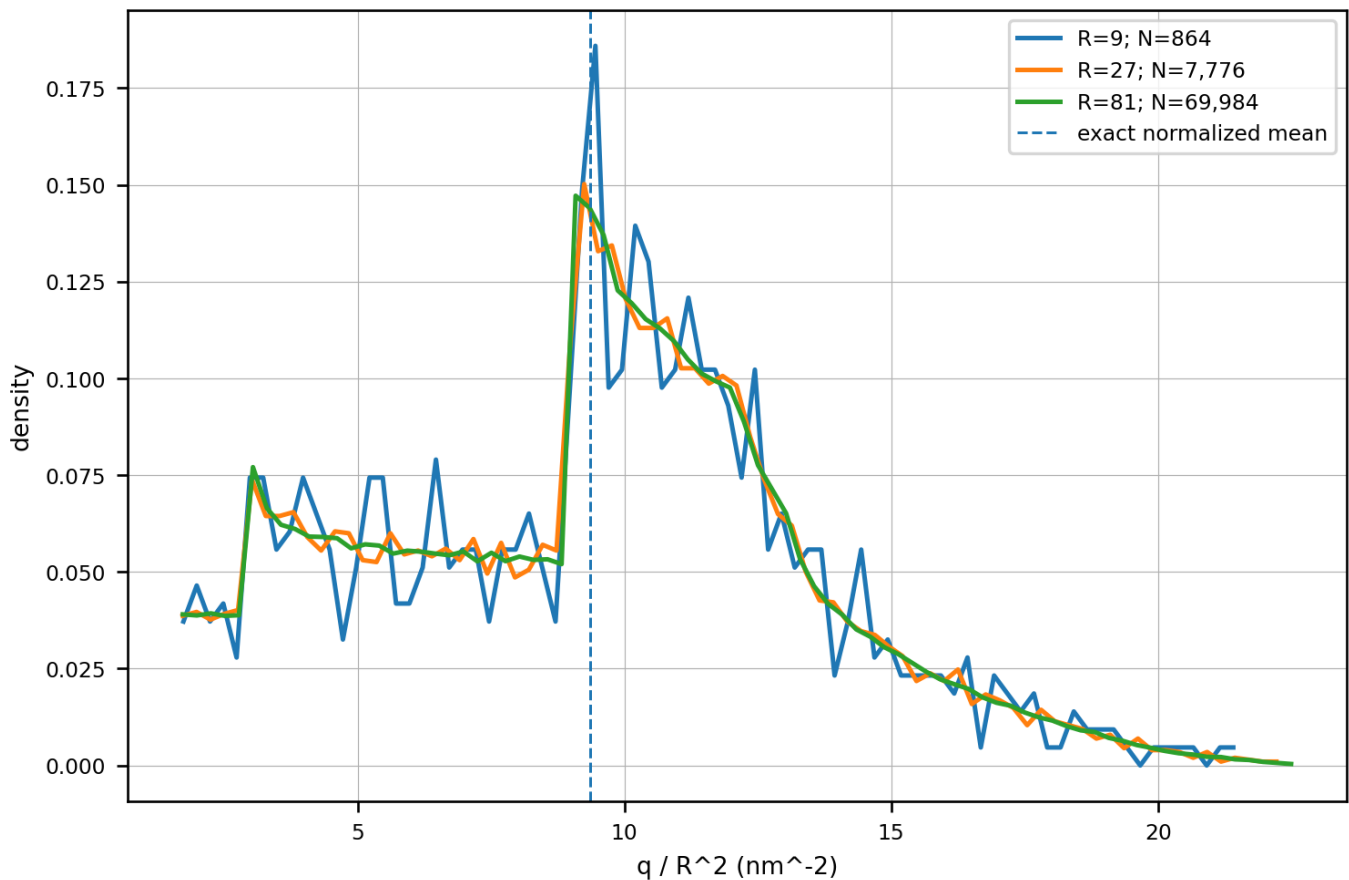


Figure 13.1. Exact normalized primitive q distributions at radii 9, 27, and 81. The number of directions increases from 864 to 7,776 to 69,984, while the normalized distribution converges toward a fixed metric-dependent form.

13.5 The 1+2+3 metric sectors

The six quadratic coordinate features separate into one trace scalar, two diagonal-traceless components, and three off-diagonal shear components. Their multiplicities form the exact 1+2+3 decomposition. Scale growth adds observations to these same sectors; it does not create additional independent quadratic metric coefficients.

14. Emily Nearest-Neighbor Operator Scaling

14.1 Fine refined lattice

The fine refined lattice has $n=18s+1$ nodes per axis. Open cubic nearest-neighbor adjacency has eigenvalues

$$\epsilon_{pqr} = 2[\cos(p \pi/(n+1)) + \cos(q \pi/(n+1)) + \cos(r \pi/(n+1))]$$

The extreme eigenvalues are $\pm 6 \cos(\pi/(18s+2))$ and approach ± 6 as s grows. The complement-center zero-mode family has $3n-2=54s+1$ modes, equal to the number of refined laminar sheets. The computational sweep through $s=243$ found no additional fine zero modes in this $n=18s+1$ family.

$$\begin{aligned} \text{fine nullity} &= 54s+1 \\ \text{fine rank} &= (18s+1)^3 - (54s+1) \\ \text{null fraction is } &0(s^{-2}) \end{aligned}$$

14.2 Coarse center-lattice resonances

At coarse Lucas-Supergroup resolution, $n=s$. Odd s contains the complement-center family. Additional 120-degree cosine resonances occur when the denominator $s+1$ supports the required one-third angular separation and the index range is sufficiently large. The first such even-scale resonance is $s=8$ with 12 exact zero modes.

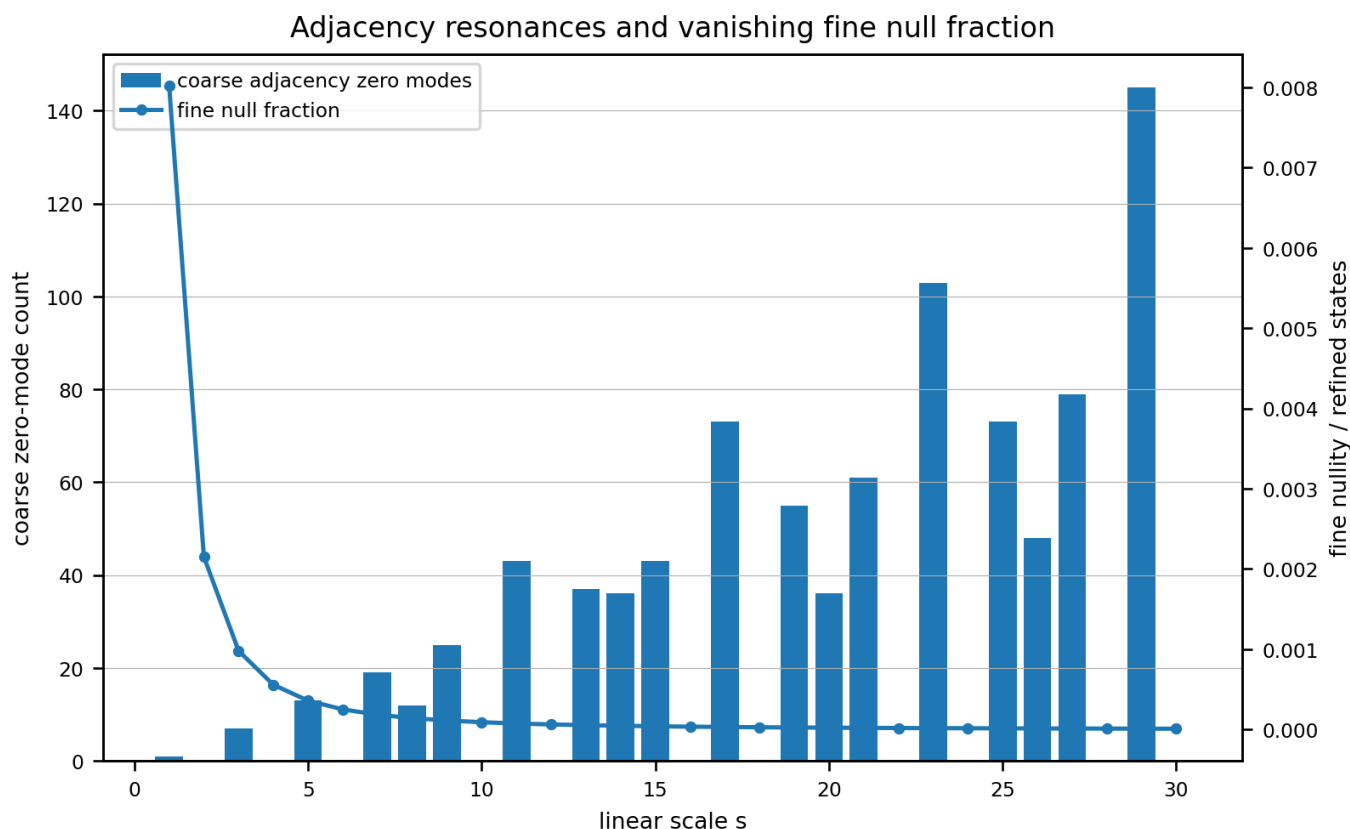


Figure 14.1. Coarse zero-mode counts for $s=1$ through 30 and fine null fraction over the same scales. Coarse arithmetic resonances appear discretely, while the fine null fraction decreases rapidly.

15. Resulting Large-Scale Morphology

The complete volume-filling morphology is now determined by exact geometry, topology, arithmetic, and spectrum. The principal properties are collected below.

Sector	Large-scale morphology
Geometric form	Fixed oblique parallelepiped under equal-count growth; no spherical limit
Boundary behavior	Quadratic boundary against cubic volume; boundary fractions decrease as inverse linear scale

Sector	Large-scale morphology
Interior behavior	Six-neighbor translational bulk first appears at $s=3$ and becomes the majority at $s=10$
Laminar behavior	$54s+1$ sheets; central populations grow as $243s^2+27s+1$
Shell behavior	Primitive directions grow as $12J_2(9s)$, with arithmetic inheritance from common divisors
Reciprocal behavior	q grows as s^2 ; q/R^2 converges to a fixed anisotropic distribution
Topological behavior	Filled body Euler characteristic 1; boundary Euler characteristic 2
Recursive behavior	$9 \times 9 \times 9$ units produce exact closure at every coarse level
Polarity behavior	Every ternary digit contributes one seven-sheet parity factor
Interface behavior	Local 3:1:1 decomposition and exact $1/9$ decay across closure depth
Closure family	21 unordered integer cuboids, 106 oriented physical morphologies, unique isotropic minimum at $9 \times 9 \times 9$

The Lucas Supergroup is therefore not a terminal 729-Cell3 object. It is the unique isotropic kernel and minimum-volume member of an extensible hierarchy. It generates seven-sheet quotient structures at ternary group scales, exact closure structures at powers of nine, a growing bulk medium, and increasingly dense shell and reciprocal observation systems while retaining the same locked anisotropic geometry.

16. Educational Interpretation of the Emergent Properties

16.1 Why the second-order closure is structurally new

A larger block ordinarily has more units but no new defining identity. The second-order Lucas closure is different because the complete coarse census is identical to the original Lucas-Supergroup census. One thousand coarse vertices, 2,700 coarse edges, 2,430 coarse faces, 729 coarse volumes, 6,859 coarse barycentric states, 55 coarse sheets, 1,944 internal coarse interfaces, 1,946 coarse boundary states, and cycle rank 1,216 all reappear together. This simultaneous recurrence fixes the second-order object as a true scale copy in the cellular sense.

16.2 Why bulk emergence is not the same as closure

Closure occurs at $s=9$, while bulk majority occurs at $s=10$. The two transitions measure different properties. Closure compares boundary states with internal interfaces and is exact at one scale. Bulk majority compares six-neighbor interior objects with all boundary objects and occurs one step later. The distinction shows that topological closure and environmental dominance are separate emergent properties.

16.3 Why arithmetic enters morphology

The shell is made of integer addresses. Primitive directions are removed whenever all three address components share a divisor. As the radius acquires new prime factors, new inherited families appear. The geometry is continuous in physical space, but the admissible address morphology carries number-theoretic structure through $J_2(R)$.

16.4 Why anisotropy persists

Every scale multiplies the same basis vectors. Linear dimensions grow, but dimensionless ratios and included angles do not change. Reciprocal principal eigenvalues scale only through address magnitude, not through a changing metric. The large-volume limit is therefore a homogeneous anisotropic medium rather than an isotropic medium.

16.5 Why the hierarchy alternates between seven sheets and closure

One balanced-ternary digit per axis produces a $3 \times 3 \times 3$ seven-sheet array. Two digits per axis produce a $9 \times 9 \times 9$ Lucas closure. Adding one further digit produces a seven-sheet array of closures; adding the next produces another complete closure. The alternation is the direct consequence of the two-digit width of the Lucas Supergroup.

17. Fixed Structural Scope and Operator Extension

The authoritative definition and the present derivation fix the direct and reciprocal geometry, the 27-node scaffold, the six boundary planes, balanced-ternary composition, complete cubical f-vectors, interior and boundary topology, interface-boundary closure, laminar laws, shell counts, primitive direction counts, reciprocal q observations, principal modes, and nearest-neighbor adjacency spectra.

The separate weighted gluing law, edge-plaquette-volume action, holonomy operator, dynamical energy functional, and physical coupling operator are not specified by the source definition. The exact multiscale census derived here supplies the complete geometric and combinatorial domain on which those operators can be defined without ambiguity. In particular, it fixes every interface population, boundary population, hierarchy level, shell address, sheet grade, and reciprocal observation required by a subsequent dynamical construction.

18. Conclusions

Systematic volume filling with complete Lucas Supergroups reveals a rigid multiscale morphology. The general aligned assembly is exactly solvable for arbitrary a, b, c . Compact cubic growth preserves the oblique physical shape, drives a boundary-to-bulk transition, and produces exact cellular closure at $s=9$. The same closure repeats recursively at every power-of-nine hierarchy level.

The $9 \times 9 \times 9$ closure is the unique isotropic and minimum-volume member of a finite family of 21 unordered integer closures. All members share universal normalized interface, boundary, exposure, cycle-rank, and coordination laws, while their physical surface areas differ under orientation because the locked metric distinguishes the three channels.

Laminar sheets, primitive shell directions, reciprocal q spectra, and adjacency modes each exhibit their own scaling laws. Together these properties define the transition from a single exact object to a filled anisotropic field volume. The extension retains every foundational invariant of Cell3 and the Lucas Supergroup while adding bulk, recursion, arithmetic inheritance, and multiscale closure as new emergent properties.

Appendix A. Complete Locked Basis and Metric Ledger

Item	Value
B row 1	0.3275012420558906, 0.0353307758438284, -0.01012931762672882
B row 2	-0.05489700788400644, 0.2073198504401918, -0.3897877023662599
B row 3	-0.03748750289472991, -0.5474010853885081, -0.6897344083672676
M row 1	0.1116760578960501, 0.02071033328060581, 0.04393723509621977
M row 2	0.02071033328060581, 0.343877732392787, 0.2963927589615229
M row 3	0.04393723509621977, 0.2963927589615229, 0.6277706100772958
R row 1	9.208025529457549, 0.001534802428192545, -0.645188035699939
R row 2	0.001534802428192587, 4.903398312367987, -2.315175584416389
R row 3	-0.6451880356999389, -2.315175584416389, 2.731171274711695
Edge lengths (nm)	0.3341796790591105, 0.5864108904111408, 0.7923197650426852
Included angles (deg)	ab=83.93348301231273; ac=80.44813001833791; bc=50.36323181375618
Face areas (nm ²)	ab=0.1948691654604841; ac=0.2611062357902053; bc=0.3578095391285615
Cell3 volume (nm ³)	0.1179148919117446
Body diagonal (nm)	1.343653621675925

Appendix B. Complete 27-Node Cell3 Ledger

ID	Address	lambda	chi	Class	ICRS position (nm)
N01	(-1, -1, -1)	-3	-1	corner	(-0.176351350136495, 0.118682429905037, 0.637311498325253)
N02	(-1, -1, 0)	-2	1	edge center	(-0.181416008949859, -0.0762114212780927, 0.292444294141619)
N03	(-1, -1, 1)	-1	-1	corner	(-0.186480667763224, -0.271105272461223, -0.0524229100420148)
N04	(-1, 0, -1)	-2	1	edge center	(-0.158685962214581, 0.222342355125133, 0.363610955630999)
N05	(-1, 0, 0)	-1	-1	face center	(-0.163750621027945, 0.0274485039420032, 0.018743751447365)
N06	(-1, 0, 1)	0	1	edge center	(-0.16881527984131, -0.167445347241127, -0.326123452736269)
N07	(-1, 1, -1)	-1	-1	corner	(-0.141020574292667, 0.326002280345229, 0.0899104129367447)
N08	(-1, 1, 0)	0	1	edge center	(-0.146085233106031, 0.131108429162099, -0.254956791246889)
N09	(-1, 1, 1)	1	-1	corner	(-0.151149891919396, -0.0637854220210308, -0.599823995430523)
N10	(0, -1, -1)	-2	1	edge center	(-0.0126007291085498, 0.091233925963034, 0.618567746877888)
N11	(0, -1, 0)	-1	-1	face center	(-0.0176653879219142, -0.103659925220096, 0.273700542694254)
N12	(0, -1, 1)	0	1	edge center	(-0.0227300467352786, -0.298553776403226, -0.0711666614893798)
N13	(0, 0, -1)	-1	-1	face center	(0.00506465881336441, 0.19489385118313, 0.344867204183634)
N14	(0, 0, 0)	0	1	body center	(0, 0, 0)
N15	(0, 0, 1)	1	-1	face center	(-0.00506465881336441, -0.19489385118313, -0.344867204183634)
N16	(0, 1, -1)	0	1	edge center	(0.0227300467352786, 0.298553776403226, 0.0711666614893798)
N17	(0, 1, 0)	1	-1	face center	(0.0176653879219142, 0.103659925220096, -0.273700542694254)
N18	(0, 1, 1)	2	1	edge center	(0.0126007291085498, -0.091233925963034, -0.618567746877888)
N19	(1, -1, -1)	-1	-1	corner	(0.151149891919396, 0.0637854220210308, 0.599823995430523)
N20	(1, -1, 0)	0	1	edge center	(0.146085233106031, -0.131108429162099, 0.254956791246889)
N21	(1, -1, 1)	1	-1	corner	(0.141020574292667, -0.326002280345229, -0.0899104129367447)
N22	(1, 0, -1)	0	1	edge center	(0.16881527984131, 0.167445347241127, 0.326123452736269)
N23	(1, 0, 0)	1	-1	face center	(0.163750621027945, -0.0274485039420032, -0.018743751447365)
N24	(1, 0, 1)	2	1	edge center	(0.158685962214581, -0.222342355125133, -0.363610955630999)
N25	(1, 1, -1)	1	-1	corner	(0.186480667763224, 0.271105272461223, 0.0524229100420148)
N26	(1, 1, 0)	2	1	edge center	(0.181416008949859, 0.0762114212780927, -0.292444294141619)
N27	(1, 1, 1)	3	-1	corner	(0.176351350136495, -0.118682429905037, -0.637311498325253)

Appendix C. Six Boundary Planes and Node Membership

Face	Family	Primitive coordinate	Outward normal (ICRS)	Area (nm ²)	Spacing (nm)	Node IDs
a=-1	bc	-0.5	(0.995965190309, -0.083602162507, 0.032619290564)	0.357809539129	0.329546529695	N01, N02, N03, N04, N05, N06, N07, N08, N09
a=+1	bc	+0.5	(-0.995965190309, 0.083602162507, -0.032619290564)	0.357809539129	0.329546529695	N19, N20, N21, N22, N23, N24, N25, N26, N27
b=-1	ac	-0.5	(0.089052594102, 0.866576769291, -0.491033948326)	0.261106235790	0.451597379721	N01, N02, N03, N10, N11, N12, N19, N20, N21
b=+1	ac	+0.5	(-0.089052594102, -0.866576769291, 0.491033948326)	0.261106235790	0.451597379721	N07, N08, N09, N16, N17, N18, N25, N26, N27
c=-1	ab	-0.5	(-0.194092200808, -0.913177168821, -0.358379234791)	0.194869165460	0.605097741519	N01, N04, N07, N10, N13, N16, N19, N22, N25
c=+1	ab	+0.5	(0.194092200808, 0.913177168821, 0.358379234791)	0.194869165460	0.605097741519	N03, N06, N09, N12, N15, N18, N21, N24, N27

Appendix D. Cubic Sweep s=1 Through s=81

The machine-readable companion extends this table through s=243 and retains every exact count and derived scalar.

s	SG	Cell3	Bulk	Bulk frac.	Cycle	Fine boundary	Fine interfaces	Sheets	Primitive dirs.
1	1	729	0	0.000000	0	1,946	1,944	55	864
2	8	5,832	0	0.000000	5	7,778	16,524	109	2,592
3	27	19,683	1	0.037037	28	17,498	56,862	163	7,776
4	64	46,656	8	0.125000	81	31,106	136,080	217	10,368
5	125	91,125	27	0.216000	176	48,602	267,300	271	20,736
6	216	157,464	64	0.296296	325	69,986	463,644	325	23,328
7	343	250,047	125	0.364431	540	95,258	738,234	379	41,472
8	512	373,248	216	0.421875	833	124,418	1,104,192	433	41,472
9	729	531,441	343	0.470508	1,216	157,466	1,574,640	487	69,984
10	1,000	729,000	512	0.512000	1,701	194,402	2,162,700	541	62,208
11	1,331	970,299	729	0.547708	2,300	235,226	2,881,494	595	103,680
12	1,728	1,259,712	1,000	0.578704	3,025	279,938	3,744,144	649	93,312
13	2,197	1,601,613	1,331	0.605826	3,888	328,538	4,763,772	703	145,152
14	2,744	2,000,376	1,728	0.629738	4,901	381,026	5,953,500	757	124,416
15	3,375	2,460,375	2,197	0.650963	6,076	437,402	7,326,450	811	186,624
16	4,096	2,985,984	2,744	0.669922	7,425	497,666	8,895,744	865	165,888
17	4,913	3,581,577	3,375	0.686953	8,960	561,818	10,674,504	919	248,832
18	5,832	4,251,528	4,096	0.702332	10,693	629,858	12,675,852	973	209,952
19	6,859	5,000,211	4,913	0.716285	12,636	701,786	14,912,910	1,027	311,040
20	8,000	5,832,000	5,832	0.729000	14,801	777,602	17,398,800	1,081	248,832
21	9,261	6,751,269	6,859	0.740633	17,200	857,306	20,146,644	1,135	373,248
22	10,648	7,762,392	8,000	0.751315	19,845	940,898	23,169,564	1,189	311,040
23	12,167	8,869,743	9,261	0.761157	22,748	1,028,378	26,480,682	1,243	456,192
24	13,824	10,077,696	10,648	0.770255	25,921	1,119,746	30,093,120	1,297	373,248
25	15,625	11,390,625	12,167	0.778688	29,376	1,215,002	34,020,000	1,351	518,400
26	17,576	12,812,904	13,824	0.786527	33,125	1,314,146	38,274,444	1,405	435,456
27	19,683	14,348,907	15,625	0.793832	37,180	1,417,178	42,869,574	1,459	629,856
28	21,952	16,003,008	17,576	0.800656	41,553	1,524,098	47,818,512	1,513	497,664
29	24,389	17,779,581	19,683	0.807044	46,256	1,634,906	53,134,380	1,567	725,760
30	27,000	19,683,000	21,952	0.813037	51,301	1,749,602	58,830,300	1,621	559,872
31	29,791	21,717,639	24,389	0.818670	56,700	1,868,186	64,919,394	1,675	829,440

s	SG	Cell3	Bulk	Bulk frac.	Cycle	Fine boundary	Fine interfaces	Sheets	Primitive dirs.
32	32,768	23,887,872	27,000	0.823975	62,465	1,990,658	71,414,784	1,729	663,552
33	35,937	26,198,073	29,791	0.828978	68,608	2,117,018	78,329,592	1,783	933,120
34	39,304	28,652,616	32,768	0.833706	75,141	2,247,266	85,676,940	1,837	746,496
35	42,875	31,255,875	35,937	0.838181	82,076	2,381,402	93,469,950	1,891	995,328
36	46,656	34,012,224	39,304	0.842421	89,425	2,519,426	101,721,744	1,945	839,808
37	50,653	36,926,037	42,875	0.846445	97,200	2,661,338	110,445,444	1,999	1,181,952
38	54,872	40,001,688	46,656	0.850270	105,413	2,807,138	119,654,172	2,053	933,120
39	59,319	43,243,551	50,653	0.853909	114,076	2,956,826	129,361,050	2,107	1,306,368
40	64,000	46,656,000	54,872	0.857375	123,201	3,110,402	139,579,200	2,161	995,328
41	68,921	50,243,409	59,319	0.860681	132,800	3,267,866	150,321,744	2,215	1,451,520
42	74,088	54,010,152	64,000	0.863838	142,885	3,429,218	161,601,804	2,269	1,119,744
43	79,507	57,960,603	68,921	0.866854	153,468	3,594,458	173,432,502	2,323	1,596,672
44	85,184	62,099,136	74,088	0.869741	164,561	3,763,586	185,826,960	2,377	1,244,160
45	91,125	66,430,125	79,507	0.872505	176,176	3,936,602	198,798,300	2,431	1,679,616
46	97,336	70,957,944	85,184	0.875154	188,325	4,113,506	212,359,644	2,485	1,368,576
47	103,823	75,686,967	91,125	0.877696	201,020	4,294,298	226,524,114	2,539	1,907,712
48	110,592	80,621,568	97,336	0.880136	214,273	4,478,978	241,304,832	2,593	1,492,992
49	117,649	85,766,121	103,823	0.882481	228,096	4,667,546	256,714,920	2,647	2,032,128
50	125,000	91,125,000	110,592	0.884736	242,501	4,860,002	272,767,500	2,701	1,555,200
51	132,651	96,702,579	117,649	0.886906	257,500	5,056,346	289,475,694	2,755	2,239,488
52	140,608	102,503,232	125,000	0.888996	273,105	5,256,578	306,852,624	2,809	1,741,824
53	148,877	108,531,333	132,651	0.891011	289,328	5,460,698	324,911,412	2,863	2,426,112
54	157,464	114,791,256	140,608	0.892953	306,181	5,668,706	343,665,180	2,917	1,889,568
55	166,375	121,287,375	148,877	0.894828	323,676	5,880,602	363,127,050	2,971	2,488,320
56	175,616	128,024,064	157,464	0.896638	341,825	6,096,386	383,310,144	3,025	1,990,656
57	185,193	135,005,697	166,375	0.898387	360,640	6,316,058	404,227,584	3,079	2,799,360
58	195,112	142,236,648	175,616	0.900078	380,133	6,539,618	425,892,492	3,133	2,177,280
59	205,379	149,721,291	185,193	0.901713	400,316	6,767,066	448,317,990	3,187	3,006,720
60	216,000	157,464,000	195,112	0.903296	421,201	6,998,402	471,517,200	3,241	2,239,488
61	226,981	165,469,149	205,379	0.904829	442,800	7,233,626	495,503,244	3,295	3,214,080
62	238,328	173,741,112	216,000	0.906314	465,125	7,472,738	520,289,244	3,349	2,488,320
63	250,047	182,284,263	226,981	0.907753	488,188	7,715,738	545,888,322	3,403	3,359,232
64	262,144	191,102,976	238,328	0.909149	512,001	7,962,626	572,313,600	3,457	2,654,208
65	274,625	200,201,625	250,047	0.910503	536,576	8,213,402	599,578,200	3,511	3,483,648
66	287,496	209,584,584	262,144	0.911818	561,925	8,468,066	627,695,244	3,565	2,799,360
67	300,763	219,256,227	274,625	0.913094	588,060	8,726,618	656,677,854	3,619	3,877,632
68	314,432	229,220,928	287,496	0.914334	614,993	8,989,058	686,539,152	3,673	2,985,984
69	328,509	239,483,061	300,763	0.915540	642,736	9,255,386	717,292,260	3,727	4,105,728
70	343,000	250,047,000	314,432	0.916711	671,301	9,525,602	748,950,300	3,781	2,985,984
71	357,911	260,917,119	328,509	0.917851	700,700	9,799,706	781,526,394	3,835	4,354,560
72	373,248	272,097,792	343,000	0.918960	730,945	10,077,698	815,033,664	3,889	3,359,232
73	389,017	283,593,393	357,911	0.920039	762,048	10,359,578	849,485,232	3,943	4,603,392
74	405,224	295,408,296	373,248	0.921091	794,021	10,645,346	884,894,220	3,997	3,545,856
75	421,875	307,546,875	389,017	0.922114	826,876	10,935,002	921,273,750	4,051	4,665,600
76	438,976	320,013,504	405,224	0.923112	860,625	11,228,546	958,636,944	4,105	3,732,480
77	456,533	332,812,557	421,875	0.924084	895,280	11,525,978	996,996,924	4,159	4,976,640
78	474,552	345,948,408	438,976	0.925032	930,853	11,827,298	1,036,366,812	4,213	3,919,104
79	493,039	359,425,431	456,533	0.925957	967,356	12,132,506	1,076,759,730	4,267	5,391,360

s	SG	Cell3	Bulk	Bulk frac.	Cycle	Fine boundary	Fine interfaces	Sheets	Primitive dirs.
80	512,000	373,248,000	474,552	0.926859	1,004,801	12,441,602	1,118,188,800	4,321	3,981,312
81	531,441	387,420,489	493,039	0.927740	1,043,200	12,754,586	1,160,667,144	4,375	5,668,704

Appendix E. Complete Unordered Rectangular Closure Ledger

a	b	c	N=abc	I=8N/3	B=8N/3+2	E=2N/3	cycle=5N/3+1
4	13	156	8,112	21,632	21,634	5,408	13,521
4	14	84	4,704	12,544	12,546	3,136	7,841
4	15	60	3,600	9,600	9,602	2,400	6,001
4	16	48	3,072	8,192	8,194	2,048	5,121
4	18	36	2,592	6,912	6,914	1,728	4,321
4	20	30	2,400	6,400	6,402	1,600	4,001
4	21	28	2,352	6,272	6,274	1,568	3,921
4	24	24	2,304	6,144	6,146	1,536	3,841
5	8	120	4,800	12,800	12,802	3,200	8,001
5	9	45	2,025	5,400	5,402	1,350	3,376
5	10	30	1,500	4,000	4,002	1,000	2,501
5	12	20	1,200	3,200	3,202	800	2,001
5	15	15	1,125	3,000	3,002	750	1,876
6	7	42	1,764	4,704	4,706	1,176	2,941
6	8	24	1,152	3,072	3,074	768	1,921
6	9	18	972	2,592	2,594	648	1,621
6	10	15	900	2,400	2,402	600	1,501
6	12	12	864	2,304	2,306	576	1,441
7	7	21	1,029	2,744	2,746	686	1,716
8	8	12	768	2,048	2,050	512	1,281
9	9	9	729	1,944	1,946	486	1,216

Appendix F. Recursive Closure Ledger

L	Side Cell3	Cell3 volumes	Lucas SG	Refined states	Internal interfaces	Boundary states	Exposed faces	Level contributions
1	9	729	1	6,859	1,944	1,946	486	1,944
2	81	531,441	729	4,330,747	1,574,640	157,466	39,366	1,417,176, 157,464
3	729	387,420,489	531,441	3,105,745,579	1,160,667,144	12,754,586	3,188,646	1,033,121,304, 114,791,256, 12,754,584
4	6,561	282,429,536,481	387,420,489	2,259,952,891,867	847,159,469,280	1,033,121,306	258,280,326	753,145,430,616, 83,682,825,624, 9,298,091,736, 1,033,121,304
5	59,049	205,891,132,094,649	282,429,536,481	1,647,170,898,524,299	617,662,935,930,744	83,682,825,626	20,920,706,406	549,043,018,919,064, 61,004,779,879,896, 6,778,308,875,544, 753,145,430,616, 83,682,825,624

Appendix G. Reproducibility Equations

Direct metric: $M = B^T B$

Reciprocal metric: $R = M^{-1}$

Cell3 local embedding: $r(u) = B u/2$

Supergroup center embedding: $r = 9B g$

General outer basis: $B_{out} = 9B \text{ diag}(a, b, c)$

General f-polynomial: $[(a+1)+at][(b+1)+bt][(c+1)+ct]$

Coarse boundary: $8(ab+ac+bc)+2$

Coarse interfaces: $3abc-(ab+ac+bc)$

Rectangular closure: $1/a+1/b+1/c=1/3$

Fine sheet radius: $R=9s$

Sheet count: $54s+1$

Primitive directions: $12J2(9s)$

Reciprocal scalar: $q=u^T R u$

Mean q: $(5/9)R^2 \text{ trace}(R)$

Fine Emily size: $n=18s+1$

Fine Emily extremes: $\pm 6\cos(\pi/(18s+2))$

Appendix H. Source and Archival Record

Field	Record
Authoritative source title	K-Field Lucas Supergroup Extracted Definition
Authoritative source schema	K-Field.lucas_supergroup.extracted_definition.v1
Authoritative source epoch	1785238606
Authoritative source file	K-Field_Lucas_Supergroup_extracted_definition_1785238606(1).json
Authoritative source SHA-256	79b8aadb50be81da2569a901063db166fd5710377d644412638b25ffacd4526a
Report title	K-Field Lucas Supergroup Large-Scale Morphology
Report epoch	1785242330
Report generated UTC	2026-07-28T12:38:50Z
Report PDF	K-Field_Lucas_Supergroup_large_scale_morphology1785242330.pdf
Machine-readable companion	K-Field_Lucas_Supergroup_large_scale_morphology1785242330.json
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx, LLC
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation library	Python ReportLab
PDF fonts	Helvetica; Helvetica-Bold; Times-Roman; Courier
Layout controls	Dynamic table widths; Paragraph-wrapped text cells; repeatRows=1 for long tables; 0.68 inch margins

P. Dwayne Esterline

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